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Calculus of Variation

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Calculus of Variation

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CHAPTER 1

CALCULUS OF VARIATION

The calculus of variations has its origin in the generalization of the lementary theory of maxima and minima of function of a single or more variables. Its object is to find extrema or stationary values of functionals. By a functional we mean a quantity whose values are determined by one or several functions.

The aim of calculus of variations is to explore methods for finding the maximum or minimum of a functional defined over a class of functions.

1.1. Variation of a Functional

Let $F = F(x, y, y')$ be a functional, which is a function of y, y', x (independent & suppose to be fixed).

Let $y = y(x)$ is changed into $y + \varepsilon\eta(x)$

Where ε is small parameter and $\eta(x)$ is an arbitrary function of x. Then

$y' \to y' + \in \eta'(x)$.

$\therefore F = F(x, y + \in \eta, y' + \in \eta')$ and so the change in F is

$$\Delta F = F(x, y + \varepsilon\eta, y' + \varepsilon\eta') - F(x, y, y') \quad \ldots(1)$$

By Taylor's series expansion

$$F(x, y + \varepsilon\eta, y' + \varepsilon\eta') = F(x, y, y') + \left(\varepsilon\eta\frac{\partial}{\partial y} + \varepsilon\eta'\frac{\partial}{\partial y'}\right)F + \frac{1}{2!}\left(\varepsilon\eta\frac{\partial}{\partial y} + \varepsilon\eta'\frac{\partial}{\partial y'}\right)^2 F$$

$$+ \text{ higher terms of } \in \quad \ldots(2)$$

By (1) & (2), we get

$$\Delta F = \left(\varepsilon\eta\frac{\partial F}{\partial y} + \varepsilon\eta'\frac{\partial F}{\partial y'}\right) + \frac{\varepsilon^2}{2!}\left(\eta^2\frac{\partial^2 F}{\partial y^2} + 2\eta\eta'\frac{\partial^2 F}{\partial y \partial y'} + \eta'^2\frac{\partial^2 F}{\partial y'^2}\right) + 0(\varepsilon^2)$$

$$\ldots(3)$$

The linear part in (3) is called first variation in F (or simply variations) and the term of $\varepsilon^2, \varepsilon^3$ -------- are called the second, third, variation etc. So the first variation is denoted by

$$\delta F = \frac{\partial F}{\partial y}\varepsilon\eta + \frac{\partial F}{\partial y'}\varepsilon\eta' \quad \ldots(4)$$

y is changed into $y + \varepsilon\eta$, so the variation in y is $\delta y = \varepsilon\eta$ and similarly $\delta y' = \varepsilon\eta'$.

Now, we get

$$\delta F = \frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y'$$

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We know that for a function:

$$F = F(x, y, y')$$

$$dF = \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy + \frac{\partial F}{\partial y'}dy'$$

Since x is fixed $dx = 0$

$$dF = \frac{\partial F}{\partial y}dy + \frac{\partial F}{\partial y'}dy' \qquad \ldots(6)$$

Note: The operations δ and $\frac{d}{dx}$ are commutative.

We have

$$y' \to y' + \varepsilon\eta'$$

$$\delta y = \varepsilon\eta$$

$$\delta y' = \varepsilon\eta'$$

Then the variation in $\frac{dy}{dx}$, is

$$\delta\left(\frac{dy}{dx}\right) = \varepsilon\frac{d\eta}{dx} \qquad \ldots(*)$$

$$\frac{d}{dx}(\delta y) = \frac{d}{dx}(\varepsilon\eta) = \varepsilon\frac{d\eta}{dx} = \delta\left(\frac{dy}{dx}\right)$$ (By *) Hence $\frac{d}{dx}(\delta y) = \delta\frac{dy}{dx}$

$$\Rightarrow \frac{d}{dx}\delta = \delta\frac{d}{dx}.$$

1.2. The Simplest Variational Problem

To determine a function $y(x)$, which makes the integral (functional)

$$I = \int_{x_1}^{x_2} F(x, y, y')\,dx \qquad \ldots(1)$$

stationary (to have prescribed boundary conditions.

$$y(x_1) = y_1, \quad y(x_2) = y_2 \qquad \ldots(2)$$

is known as the "simplest variational problem".

Let $y = y(x)$ be our actual function which makes (1) stationary and satisfies conditions (2).

Let $\eta(x)$ be an arbitrary function of x (vanishing at x_1 and x_2) i.e.

$$\eta(x_1) = \eta(x_2) = 0$$

Then $y + \varepsilon\eta$ will also make (1) to be stationary & satisfy the (2).

Let $y' + \varepsilon\eta'$ be the change of y' then our function (1) becomes

$$I(\varepsilon) = \int_{x_1}^{x_2} F(x, y + \varepsilon\eta, y' + \varepsilon\eta')dx \qquad \ldots(3)$$

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We say that $I(\varepsilon)$ is same as I, when $\varepsilon=0$

For (1) to be stationary, it is enough to find that (2) is stationary at $\varepsilon=0$.

We know that $\dfrac{dI(\varepsilon)}{d\varepsilon}=0$, $F=F(x,y,y')$

Let us put $F_\varepsilon=F_\varepsilon(x,\, y+\varepsilon\eta, y'+\varepsilon\eta')$

Then $F_\varepsilon\to F$ as $\varepsilon\to 0$

So (3) can be written as

$$I(\varepsilon)=\int_{x_1}^{x_2}F_\varepsilon dx \qquad \ldots(4)$$

Since the limits in (3) and (4) are independent of $\in$, we get

$$\frac{dI(\varepsilon)}{d\varepsilon}=\int_{x_1}^{x_2}\frac{dF_\varepsilon}{d\varepsilon}dx \qquad \ldots(5)$$

Since $F_\varepsilon=F_\varepsilon(x,y+\varepsilon\eta,y'+\varepsilon\eta')$

$$\frac{dF_\varepsilon}{d\varepsilon}=\eta\frac{\partial F_\varepsilon}{\partial y}+\eta'\frac{\partial F_\varepsilon}{\partial y'}$$

$$0=\frac{dI(\varepsilon)}{d\varepsilon}=\int_{x_1}^{x_2}\left[\eta\frac{\partial F_\varepsilon}{\partial y}+\eta'\frac{\partial F_\varepsilon}{\partial y'}\right]dx \quad \text{(by 5)}$$

For $I(\varepsilon)$ to be stationary

$$\int_{x_1}^{x_2}\left[\eta\frac{\partial F_\varepsilon}{\partial y}+\eta'\frac{\partial F_\varepsilon}{\partial y'}\right]dx=0$$

Hence I in (1) is stationary if

$$\int_{x_1}^{x_2}\left[\eta\frac{\partial F}{\partial y}+\eta'\frac{\partial F}{\partial y'}\right]dx=0 \text{ Since } F_\varepsilon\to F \text{ as } \varepsilon\to 0$$

$$\Rightarrow \int_{x_1}^{x_2}\eta\frac{\partial F}{\partial y}dx+\int_{x_1}^{x_2}\eta'\frac{\partial F}{\partial y'}dx=0$$

$$\int_{x_1}^{x_2}\eta(x)\frac{\partial F}{\partial y}dx+\left[\frac{\partial F}{\partial y'}\eta(x)\right]_{x_1}^{x_2}-\int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\eta(x)dx=0$$

$$(\eta(x_1)=\eta(x_2)=0)$$

$$\Rightarrow \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]\eta(x)dx=0 \qquad \ldots(6)$$

If (6) holds for all $n(x)$

We must have

$$y=y(x) \text{ satisfying } \frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0 \qquad \textbf{(Euler's Equation)} \qquad (A)$$

Put Your Own Notes

Note: Depending on the boundary condition, the solution of ODE can be unique, infinite or no solution. Hence extremal of the functional can be unique infinite or no extremal.

Example: $I\left[y(x)\right]=\int_0^\pi\left(y^2-y'^2\right)dx \quad y(0)=0,\ y(\pi)=0$

Solution: $F(x,y,y')=y^2-y'^2$ ...(1)

$$\frac{\partial F}{\partial y}=2y$$

$$\frac{\partial F}{\partial y'}=-2y'$$

Using Euler's Method, we have

$$\frac{\partial F}{\partial y}-\frac{d}{dx}(-2y')=0$$

$$2y+2y''=0$$

The auxiliary equation

$$y''+y=0$$

$$\therefore y=C_1\cos x+C_2\sin x \qquad \dots(2)$$

At $y(0)=0,\ y(\pi)=0$

$$y(0)=C_1+C_2\times 0=0$$

$$C_1=0$$

$$y(x)=C_2\sin x$$

Hence the required solution is infinite extremal

Note: Assuming F is three times differentiable and $y(x)$ is twice differentiable and its called admissible function (Admit same condition).

1.3. Other Forms of Euler's Equation

(a) We have $F=F(x,y,y')$

If $\dfrac{\partial F}{\partial y'}=g(x,y,y')$

Then $\dfrac{d}{dx}\left(\dfrac{\partial F}{\partial y'}\right)=\dfrac{dg}{dx}=\dfrac{\partial g}{\partial x}+\dfrac{\partial g}{\partial y}.\dfrac{dy}{dx}+\dfrac{\partial g}{\partial y'}\dfrac{dy'}{dx}$

$$=\frac{\partial}{\partial x}\left(\frac{\partial F}{\partial y'}\right)+\frac{\partial}{\partial y}\left(\frac{\partial F}{\partial y'}\right)y'+\frac{\partial}{\partial y'}\left(\frac{\partial F}{\partial y'}\right)y''=\frac{\partial^2F}{\partial x\partial y'}+\frac{\partial^2F}{\partial y\partial y'}y'+\frac{\partial^2F}{\partial y'^2}y''$$

Putting this in (A)

$$\frac{\partial^2F}{\partial y'^2}y''+\frac{\partial^2F}{\partial y\partial y'}y'+\frac{\partial^2F}{\partial x\partial y'}-\frac{\partial F}{\partial y}=0 \qquad (B)$$

Put Your Own Notes

(b) $F = F(x, y, y')$, then

$$\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} + \frac{\partial F}{\partial y'}\frac{dy'}{dx}$$

$$\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}y' + \frac{\partial F}{\partial y'}y''$$

$$\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}y' + \frac{\partial F}{\partial y'}y'' \quad \ldots(1)$$

$$\frac{\partial F}{\partial y}y' = \frac{dF}{dx} - \frac{\partial F}{\partial x} - \frac{\partial F}{\partial y'}y''$$

Since $\frac{d}{dx}\left(\frac{\partial F}{\partial y'}y'\right) = \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)y' + \left(\frac{\partial F}{\partial y'}\right)y''$ $\quad \ldots(2)$

By (1) & (2)

$$\frac{dF}{dx} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}y'\right) = \frac{\partial F}{\partial x} + \left[\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]y' = \frac{\partial F}{\partial x} \quad \text{(By equation}(A))$$

$$\frac{dF}{dx} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}y'\right) = \frac{\partial F}{\partial x}$$

i.e., $\frac{d}{dx}\left[F - y'\frac{\partial F}{\partial y'}\right] = \frac{\partial F}{\partial x}$ $\quad (C)$

1.4. Commutativity of Variation and Integration

$$I = \int_{x_1}^{x_2} F(x, y, y')dx \quad \ldots(1)$$

$y \to y + \varepsilon\eta$

$y' \to y' + \varepsilon\eta'$

$$I = \int_{x_1}^{x_2} F(x, y + \varepsilon\eta, y' + \varepsilon\eta')dx$$

Then the variation in the integral (1) is

$$\delta I = \int_{x_1}^{x_2} \left[F(x, y + \varepsilon\eta, y' + \varepsilon\eta') - F(x, y, y')\right]dx$$

$$= \int_{x_1}^{x_2} \left[\varepsilon\eta\frac{\partial F}{\partial y} + \varepsilon\eta'\frac{\partial F}{\partial y'} + term\ containing\ \varepsilon^2, \varepsilon^2 = 0\right]dx$$

$$\delta I = \int_{x_1}^{x_2} \left[\frac{\partial F}{\partial y}\delta y + \frac{\partial F}{\partial y'}\delta y'\right]dx$$

$$= \int_{x_1}^{x_2} \delta F\, dx$$

$$\delta\int_{x_1}^{x_2} F(x, y, y')dx = \int_{x_1}^{x_2} \delta F(x, y, y')dx$$

$$\delta\int = \int\delta.$$

Put Your Own Notes

$$y = c\sqrt{1+y'^2}$$

$$y^2 = c^2\left(1+y'^2\right) \Rightarrow 1 + y'^2 = \frac{y^2}{c^2}$$

$$\Rightarrow y'^2 = \frac{y^2 - c^2}{c^2}$$

$$\Rightarrow y' = \frac{\sqrt{-c^2+y^2}}{c}$$

$$\int \frac{1}{\sqrt{y^2-c^2}}dy = \int \frac{dx}{c}$$

$$\cos h^{-1}\frac{y}{c} = \frac{x}{c} + k$$

$$\Rightarrow \frac{y}{c} = \cosh\left(\frac{x}{c}+k\right)$$

$$\Rightarrow y = c \quad \cosh\left(\frac{x}{c}+k\right)$$

Hence the required curve is Catenary.

Problem: Find the extremals of the following functional

(i) $\int_{x_1}^{x_2} \frac{dy}{dx}\left(1+x^2\frac{dy}{dx}\right)dx$ (Hints: Solution of Euler's equation are called extremals)

(ii) $\int_{x_0}^{x_1} y'(x+y')dx$

(iii) $\int_{x_0}^{x_1}\left[y^2+\left(\frac{dy}{dx}\right)^2 - 2y\sin x\right]dx$

1.9. Brachistochron Problem

Brochistos = shortest

Chrones = time

Problem: Find the path on which a particle in the absence of friction will slides from one point to another point in the shortest time, under the action of gravity.

Solution: For our convenience, we take x – axis as downward and y-axis as horizontal to the right hand side and the origin as the starting point.

Let $Q(x_2, y_2)$ be the terminating point.

Let m be the mass of the particle and v be its velocity at point $p(x,y)$ .Then by the principle of conservation of energy

$$\frac{1}{2}mv^2 = mgh$$

Put Your Own Notes

$$v=\sqrt{2gh}$$

$$dt=\frac{ds}{\sqrt{2gx}}$$

$$dt=\frac{\sqrt{1+y'^2}}{\sqrt{2gx}}dx$$

$$\sqrt{2g}T=\int_0^{x_2}\frac{\sqrt{1+y'^2}}{\sqrt{x}}dx \qquad \ldots(1)$$

For this to be minimum, the Euler's equation is

$$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y}\right)=0$$

Where $F=\dfrac{\sqrt{1+y'^2}}{\sqrt{x}}$

$$\frac{\partial F}{\partial y}=0,\ \frac{\partial F}{\partial y'}=\frac{y'}{\sqrt{x}\sqrt{1+y'^2}}$$

$$0-\frac{d}{dx}\left(\frac{y'}{\sqrt{x}\sqrt{1+y'^2}}\right)=0$$

$$\Rightarrow\ \frac{y'}{\sqrt{x}\sqrt{1+y'^2}}=c$$

$$\Rightarrow\ y'^2=c^2x(1+y'^2)$$

$$\Rightarrow\ y'^2-c^2xy'^2=c^2x$$

$$\Rightarrow\ y'^2(1-c^2x)=c^2x$$

$$\Rightarrow\ y'=\frac{c\sqrt{x}}{\sqrt{1-c^2x}}$$

$$\Rightarrow\ y=\int_0^x\frac{c\sqrt{x}}{\sqrt{1-c^2x}}dx \qquad \ldots(2)$$

Put $x=\dfrac{1}{c^2}\sin^2\theta$

$$dx=2\frac{1}{c^2}\sin\theta\cos\theta\,d\theta$$

From (2)

$$y=\int_0^\theta\frac{c\sin\theta}{c\cos\theta}\cdot\frac{2}{c^2}\sin\theta\cos\theta\,d\theta$$

$$=\frac{1}{c^2}\int_0^\theta(1-\cos2\theta)\,d\theta$$

Put Your Own Notes

$$=\frac{1}{c^2}\left[\theta-\frac{\sin 2\theta}{2}\right]$$

$$y=y(\theta)=\frac{\theta}{c^2}-\frac{\sin 2\theta}{2c^2}$$

$$y=y(\theta)=\frac{1}{2c^2}[2\theta-\sin 2\theta] \quad \ldots(3)$$

s. to $x=x(\theta)=\frac{1}{2c^2}[1-\cos 2\theta]$ ...(4)

put $\frac{1}{2c^2}=a$, $2\theta=\phi$

Then (3) and (4) become

$$\begin{bmatrix} x=x(\varphi)=a(1-\cos\varphi) \\ y=y(\varphi)=a(\varphi-\sin\varphi) \end{bmatrix} \text{ Cycloid.}$$

Problem: On what curves can function

$$I=\int_0^1\left[\left(\frac{dy}{dx}\right)^2+12xy\right]dx \text{ with } y(0)=0 \ \& \ y(1)=1 \text{ be extremized?}$$

Sol: $F=y'^2+12xy$, $\frac{\partial F}{\partial y}=12x$, $\frac{\partial F}{\partial y'}=2y'$

Euler's equation

$$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0$$

$$12x-\frac{d}{dx}(2y')=0$$

$$\Rightarrow \frac{d}{dx}(2y')=12x$$

$$2y'=\frac{12x^2}{2}+c$$

$$y'=3x^2+c_1, \quad c_1=\frac{c}{2}$$

$$y=x^3+c_1x+c_2$$

$$y(0)=c_2=0$$

$$y(1)=1+c_1+c_2=1$$

$$\Rightarrow c_1=0$$

$$\Rightarrow y=x^3.$$

Exercise: Find the curve for which the following function be extremum.

(a) $\int_{x_0}^{x_1}\left(y^2-y'^2-2y\sin x\right)dx$

(b) $\int_1^2 \frac{x^2}{y'^2}dx,\ y(1)=1,\ y(2)=4$

(c) $\int_{x_0}^{x_1}\frac{y'^2}{x}dx.$

(d) $\int_{x_0}^{x_1}\left(y'^2+2yy'-16y^2\right)dx.$

(e) $\int_{x_0}^{x_1}\frac{\sqrt{1+y'^2}}{y}dx$

(f) $F=y^2-y'^2-2y\sin x$

Put Your Own Notes

1.10. Variation Problem Becomes Meaningless

$I=\int_{x_1}^{x_2}F(x,y,y')\,dx$

$y=y(x),\ y(x_1)=y_1,\ y(x_2)=y_2$, then the variational problem becomes meaningless.

(i) If we get the identity $0=0$

(ii) $y(x)$, but if does not satisfy the given condition

(iii) If the integrand

$$F=F(x,y,y')=M(x,y)+N(x,y)y'. \qquad \ldots(*)$$

In this case the Euler's equation is reduced to $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$

$$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0 \qquad (A)$$

By $(*)$, we get

$$\frac{\partial F}{\partial y}=\frac{\partial M}{\partial y}+y'\frac{\partial N}{\partial y}$$

$$\frac{\partial F}{\partial y'}=N(x,y)$$

So $(A)\Rightarrow \frac{\partial M}{\partial y}+y'\frac{\partial N}{\partial y}-\frac{d}{dx}N(x,y)=0$

$$\frac{\partial M}{\partial y}+y'\frac{\partial N}{\partial y}-\frac{\partial N}{\partial x}-y'\frac{\partial N}{\partial y}=0 \qquad \left(\frac{dN}{dx}=\frac{\partial N}{\partial x}+\frac{\partial N}{\partial y}\frac{dy}{dx}\right)$$

$$\Rightarrow \frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}=0$$

In this case $F(x,y,y')$ becomes exact $d\left(f(x,y)\right)$

Put Your Own Notes

$$I=\int_{x_1}^{x_2} d\left(f\left(x,y\right)\right)=0$$

We conclude that I depends only upon the end points and not on the path of integration .Therefore, finding the curve $y=y(x)$, which extremize the function I , is meaningless. So the variation problem becomes meaningless.

Example: $I=\int_{x_0}^{x_1}\left(y^2+2xy\ y'\right)dx$, $y(x_0)=y_0, y(x_1)=y_1$

$$F=y^2+\ 2xyy'$$

By Euler's equation , we get $0=0$

$$F=y^2+\ 2xyy'=M(x,y)+N(x,y)y'$$

$$\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}=0$$

$$2y=2y$$

$\Rightarrow$ variation problem becomes meaning .

$$I=\int_{x_0}^{x_1}\left(y^2+2xy\ y'\right)dx=\int_{x_0}^{x_1}d(xy^2)=xy^2\Big|_{x_0}^{x_1}$$

$$=x_1y_1^2-x_0y_0^2$$

The integral is independent of the path so we cannot extremize.

Example: Test for on extremum for the functional

$$I=\int_0^2\left(xy+y^2-2y^2y'\right)dx$$

$$y(0)=1,\ \ y(1)=2$$

$$F=xy+y^2-2y^2y'$$

$$\frac{\partial F}{\partial y}=x+2y-\ 4yy',\quad \frac{\partial F}{\partial y'}=-2y^2$$

From (A)

$$x+2y-4yy'+\frac{d}{dx}\left(2y^2\right)=0$$

$$x+2y-4yy'+4yy'=0\Rightarrow x=-\ 2y,$$

$$\Rightarrow y(x)=-\frac{x}{2}$$

$$y(0)=-\frac{0}{2}=0\quad y(1)=-\frac{1}{2}$$

In this case curve function $y=-\dfrac{x}{2}$ does not satisfies the given conditions, so the variational problem becomes meaningless.

Put Your Own Notes

Exercise:

(i) Test for the extremum of the functional

$$I=\int_0^{\pi/2}\left(y'^2-y^2\right)dx$$

$$y(0)=0,\ y(\pi/2)=1$$

(ii) Extremize the following functionals.

(a) $I=\int_{x_1}^{x_2}\frac{y-xy'}{y^2}dx,\ y(x_1)=y_1\ \&\ y(x_2)=y_2$

(b) $I=\int_{x_1}^{x_2}\frac{2yx-y^2y'}{x^2}dx,\ y(x_1)=y_1\,\&\,y(x_2)=y_2$

(c) $I=\int_0^{\pi/2}\left(y'^2-y^2\right)dx,\ y(0)=0,\ y(\pi/2)=1$

(d) $I=\int_0^1\left(x'^2+1\right)dt,\ x(0)=1,\ x(1)=2$

(e) $I=\int_1^2\frac{\sqrt{1+y'^2}}{x}dx,\ y(1)=0,\ y(2)=1$

(f) $\int_{x_0}^{x_2}\left(y^2+y'^2+2ye^x\right)dx$

(g) $\int_{x_1}^{x_2}\frac{\sqrt{1+y'^2}}{x}dx$

(h) $\int_{x_0}^{x_2}(y+xy')dx$

(i) $\int_{x_0}^{x_2}\left(x^2y'^2+2y^2+2xy\right)dx$

(j) $\int_0^{\pi/2}\left[\left(\frac{dy}{dt}\right)^2-y^2+2ty\right]dt,\ y(0)=0,\ y(\pi/2)=0$

(k) $\int_0^1\left(y^2+x^2y'\right)dx,\ y(0)=0,\ y(1)=0$

(l) $\int_{x_1}^{x_2}\left(xy^2+x^2yy'\right)dx,\ y(x_1)=y_1,\ y(x_2)=y_2$

1.11. Functions Involving Higher Derivatives

$$I=\int_{x_1}^{x_2}F(x,y,y',y'')dx \quad \ldots(1)$$

We have to find $y(x)$, such that the functional (1) is stationary under the boundary condition (2) :

$$y(x_1)=y_1,\ y(x_2)=y_2,\ y'(x_1)=y_1',\ y'(x_2)=y_2' \quad \ldots(2)$$

Let $y(x)$ be the actual curve which extremize (1) under the conditions (2)

Put Your Own Notes

Let us choose an arbitrary differentiable function $\eta(x)$ such that

$$\eta(x_1)=\eta(x_2)=\eta'(x_1)=\eta'(x_2)=0$$

Let ε be a parameter then (replace y by $y+\varepsilon\eta$) in (1), then we get

$$I(\varepsilon)=\int_{x_1}^{x_2} F(x,y+\varepsilon\eta,y'+\varepsilon\eta',y''+\varepsilon\eta'') \qquad \ldots(3)$$

$$F=F(x,y,y',y'')$$

$I(\varepsilon)$ is same as I at $\varepsilon=0$

The condition for (ε) to be extremize is $\dfrac{dI(\varepsilon)}{d\varepsilon}=0$ ...(4)

By (4), since the limits of integral do not depend upon ε.

$$\frac{dI(\varepsilon)}{d\varepsilon}=\int_{x_1}^{x_2}\frac{dF_\varepsilon}{d\varepsilon}dx=0 \qquad \ldots(5)$$

Now, since $F_\varepsilon=F(x,y+\varepsilon\eta,\ y'+\varepsilon\eta',\ y''+\varepsilon\eta'')$

$$\frac{dF_\varepsilon}{d\varepsilon}=\frac{\partial F_\varepsilon}{\partial y}\eta+\frac{\partial F_\varepsilon}{\partial y'}\eta'+\frac{\partial F_\varepsilon}{\partial y''}\eta''$$

Putting this in (5)

$$\int_{x_1}^{x_2}\left[\frac{\partial F_\varepsilon}{\partial y}\eta+\frac{\partial F_\varepsilon}{\partial y'}\eta'+\frac{\partial F_\varepsilon}{\partial y''}\eta''\right]dx=0$$

Therefore for (1) to be extremum, we have

$$\int_{x_1}^{x_2}\left[\frac{\partial F_\varepsilon}{\partial y}\eta+\frac{\partial F_\varepsilon}{\partial y'}\eta'+\frac{\partial F}{\partial y''}\eta''\right]dx=0$$

$$\int_{x_1}^{x_2}\frac{\partial F}{\partial y}\eta dx+\left[\frac{\partial F}{\partial y'}\eta(x)\right]_{x_1}^{x_2}-\int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{dF}{dy'}\right)\eta dx+\left[\frac{\partial F}{\partial y''}\eta'(x)\right]_{x_1}^{x_2}-\int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{\partial F}{\partial y''}\right)\eta' dx=0$$

$$\int_{x_1}^{x_2}\frac{\partial F}{\partial y}\eta dx-\int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\eta\, dx-\left[\frac{d}{dx}\left(\frac{\partial F}{\partial y''}\right)\eta(x)\right]_{x_1}^{x_2}+\int_{x_1}^{x_2}\frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right)\eta\, dx=0$$

$$\int_{x_1}^{x_2}\frac{\partial F}{\partial y}\eta dx-\int_{x_1}^{x}\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\eta\, dx+\int_{x_1}^{x}\frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right)\eta\, dx=0$$

$$\Rightarrow \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)+\frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right)\right]\eta\, dx=0$$

This holds good, if

$$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)+\frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right)=0 \qquad \ldots(6)$$

Put Your Own Notes

General Case: If $I=\int_{x_1}^{x_2} F\left(x, y, y', y''----, y^{(n)}\right) dx$

Given $y(x_1)=y_1,\ y(x_2)=y_2$

$y'(x_1)=y_1', y(x_2)=y_2'$

$y^{(n)}(x_1)=y_1^{(n)},\ y^{(n)}(x_2)=y_2^{(n)}$

Equation analogous to equation (6), will be

$$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)+\frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right)---------+(-1)^n \frac{d^n}{dx^n}\left(\frac{\partial F}{\partial y^n}\right)=0 \qquad \ldots(7)$$

This equation is know as the Euler's Poisson equation

Exercise:

(i) Find the extremals for the following

(a) $I=\int_{x_1}^{x_2}(2xy+y''')dx$

(b) $I=\int_0^1\left(1+y''^2\right)dx$

$y(0)=0,\ y'(0)=1,\ y(1)=1,\ y'(1)=1$

(c) $I=\int_0^{\pi/2}\left(y''^2-y^2+x^2\right) dx$

$y(0)=1,\ y'(0)=0,\ y(\pi/2)=0,\ y'(\pi/2)=-1$

(d) $I=\int_{x_0}^{x_1}\left(16y^2-y''^2+x^2\right)dx$

(e) Find the curve joining points $(0,\ 0)$ & $(1,\ 0)$ for which the integral $\int_0^1 y''^2\, dx$ is a minimum

$y'(0)=a,\ y'(1)=b$

1.12. Variation Problem Involving Several Unknown Functions

$$I=\int F\left(x, y_1, y_2-y_n, y_1', ----y_n'\right)dx \qquad \ldots(1)$$

prescribed where the functions $y_1,\ y_2, —,\ y_n$ satisfy the conditions

$$\left.\begin{aligned} y_1(x_1)=y_{11}, y_2(x_1)=y_{21},\ldots\ldots, y_n(x_1)=y_{n1} \\ y_1(x_2)=y_{12}, y_2(x_2)=y_{22},\ldots\ldots, y_n(x_2)=y_{n2} \end{aligned}\right\} \qquad \ldots(2)$$

We have to find the functions

$y_1(x), y_2(x),\ldots\ldots, y_n(x)$ Which makes (1) stationary and satisfy the condition (2)

Let $\eta_1(x), \eta_2(x), ---, \eta_n(x)$ be arbitrary function vanishing at x_1 & x_2 then for some parameter $\varepsilon_i\ (i=1,\ldots\ldots\ldots=n)$, $y_i+\varepsilon_i\eta_i$ will satisfy condition and denote

Put Your Own Notes

$F_\varepsilon = F\left(x, y+\varepsilon_1\eta_1, ..., y_n+\varepsilon_n\eta_n, y_1'+\varepsilon_1\eta_1', ---, y_n'+\varepsilon_n\eta_n'\right)dx$

and $F = F\left(x, y_1, y_2, ---y_n, y_1' ----, y_n'\right)$

Then (1) becomes

$$I(\varepsilon) = \int_{x_1}^{x_2} F_\varepsilon dx \qquad ...(3)$$

We known that $F_\varepsilon \to F$ as $\varepsilon \to 0$ and I is stationary

I (ε) to be stationary the necessary condition is

$$\frac{\partial I(\varepsilon)}{\partial \varepsilon_i} = 0, \quad i = 1, 2, ...N$$

So by (3) the limits of integration are independent of ε_i's

$$\frac{\partial I(\varepsilon)}{\partial \varepsilon_i} = \int_{x_1}^{x_2} \frac{\partial F_\varepsilon}{\partial \varepsilon_i} dx = 0$$

$$F_\varepsilon = F\left(x, y_1+\varepsilon_1\eta_1, ..., y_n+\varepsilon_n\eta_n, y_1'+\varepsilon_1\eta_1', ..., y_n'+\varepsilon_n\eta_n'\right)$$

$$\frac{dF_\varepsilon}{d\varepsilon_1} = \frac{\partial F_\varepsilon}{\partial y_1}\eta_1 + \frac{\partial F_\varepsilon}{\partial y_1'}\eta_1'$$

$$\Rightarrow \int_{x_1}^{x_2}\left[\frac{\partial F_\varepsilon}{\partial y_1}\eta_1 + \frac{\partial F_\varepsilon}{\partial y_1'}\eta_1'\right]dx = 0 \qquad ...(4)$$

Therefore I to be stationary we have by (4), that

$$\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y_1}\eta_1 + \frac{\partial F}{\partial y_1'}\eta_1'\right]dx = 0$$

$$\Rightarrow \int_{x_1}^{x_2}\frac{\partial F}{\partial y_1}\eta_1 + \left[\frac{\partial F}{\partial y_1}\eta_1\right]_{x_1}^{x_2} - \int_{x_1}^{x_2}\frac{d}{dx}\eta_1 + \left(\frac{\partial F}{\partial y_1'}\right)\eta_1\, dx = 0$$

$$\Rightarrow \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y_1} - \frac{d}{dx}\eta_1 + \left(\frac{\partial F}{\partial y_1'}\right)\right]\eta_1 dx = 0$$

This holds good if

$$\frac{\partial F}{\partial y_1} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_1'}\right) = 0$$

Therefore in general we have

$$\frac{\partial F}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_i'}\right) = 0, \quad i = 1, 2, ...n$$

Example: Find the externals of

(a) $I[y(x), z(x)] = \int_0^{\pi/2}\left(y'^2 + z'^2 + 2yz\right)dx$

$y(0) = 0,\ y(\pi/2) = -1$

(b) $\int_{x_0}^{x_1}\left[2yx - 2y^2 + y'^2 - z'^2\right]dx$

Put Your Own Notes

1.13. Parametric Form of the Variational Problem

Let $x(t)$ and $y(t)$ be the parametric equation of curve

Consider

$$I=\int_{x_1}^{x_2} F(x,y,y')dx \quad (1)$$

Where $y'=\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{\dot{y}}{\dot{x}}$

So (1) becomes

$$I=\int F\left(x,y,\frac{\dot{y}}{\dot{x}}\right)\dot{x}\,dt \qquad \ldots(2)$$

Put $g(x,y,\dot{x},\dot{y})=F\left(x,y,\frac{\dot{y}}{\dot{x}}\right)\dot{x}$

Then (2) becomes

$$I=\int_{t}^{t_2} g(x,y,\dot{x},\dot{y})dt \qquad \ldots(3)$$

Therefore the required Euler's equations will be

$$\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)=0 \quad \text{and} \quad \frac{\partial g}{\partial y}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{y}}\right)=0$$

Example: Find the extremum of

$$A=\int_{t_1}^{t_2}(x\dot{y}-\dot{x}y)dt$$

$$g=x\dot{y}-\dot{x}y$$

$$\frac{\partial g}{\partial x}-\frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right)=0,\quad \frac{\partial g}{\partial x}=\dot{y},\quad \frac{\partial g}{\partial \dot{x}}=-y,\quad \frac{\partial g}{\partial \dot{y}}=x$$

By Euler's equation

$$\dot{y}-\frac{d}{dt}(-y)=0 \Rightarrow \dot{y}=0$$

$$\Rightarrow \quad y=y(t)=A \;\&\; x=x(t)=B$$

1.14. Isoperimetric Problems

Suppose we have to extremize the functional

$$I=\int_{x_1}^{x_2} F(x,y,y')dx \qquad \ldots(1)$$

With the prescribed conditions

$y(x_1)=y_1,\ \ y(x_2)=y_2$ and with some constraint(s) $\qquad \ldots(2)$

$$\int_{x_1}^{x_2} G(x,y,y')dx=k \text{ (Constant)} \qquad \ldots(3)$$

k is a prescribed constant, such problems in v. p. are known as the Isoperimetric problems.

Put Your Own Notes

We have to find analogous of Euler's equations for I to extremum under (2) & (3).

We choose arbitrary differentiable function $f(x)$ & $g(x)$ with

$$f(x_1)=g(x_1)=f(x_2)=g(x_2)=0 \qquad \ldots(4)$$

Then for all value of parameters $\in$ and α, the function $y+\in f(x)+\alpha g(x)$ will satisfy (2)

Assume that α depends upon $\in$.

Therefore $\alpha=0$ if $\varepsilon=0$ replace y by

$y+\varepsilon f(x)+\alpha g(x)$ in (1)

$$I(\varepsilon)=I(\varepsilon,\alpha)=\int_{x_1}^{x_2} F(x, y+\varepsilon f+\alpha g, y'+\varepsilon f'+\varepsilon g') \qquad \ldots(5)$$

$I(\varepsilon)=I$ if $\varepsilon=0$

I is extremum whenever $I(\varepsilon,\alpha)$ is extreme at $\varepsilon=0$, i.e.

$$\frac{dI(\varepsilon)}{d\varepsilon}=0 \qquad \ldots(6)$$

$$\Rightarrow \int_{x_1}^{x_2} \frac{d}{d\varepsilon} F(x, y+\varepsilon f+\alpha g, y'+\varepsilon f'+\alpha g')\,dx=0$$

$$\Rightarrow \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}\left\{f+g\left(\frac{d\alpha}{d\varepsilon}\right)_0\right\}+\frac{\partial F}{\partial y'}\left\{f'+g'\left(\frac{d\alpha}{d\varepsilon}\right)_0\right\}\right]dx=0 \qquad \ldots(7)$$

$$\Rightarrow \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}\left\{f+g\left(\frac{d\alpha}{d\varepsilon}\right)_0\right\}\right]dx+\left[\frac{\partial F}{\partial y'}\left\{f+g\left(\frac{d\alpha}{d\varepsilon}\right)_0\right\}\right]_{x_1}^{x_2}$$

$$-\int_{x_1}^{x_2}\left[\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\left\{f+g\left(\frac{d\alpha}{d\varepsilon}\right)_0\right\}\right]dx=0$$

$$\Rightarrow \left(\frac{d\alpha}{d\varepsilon}\right)_0 = -\frac{\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right] f\,dx}{\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right] g\,dx} \qquad \ldots(8)$$

Now replacing y by $y+\varepsilon f+\alpha g$ in (3) we get

$$\int_{x_1}^{x_2} G(x, y+\varepsilon f+\alpha g, y'+\varepsilon f'+\alpha g')\,dx=k$$

Let $J(\varepsilon,\alpha)=\int_{x_1}^{x_2} G(x, y+\varepsilon f+\alpha g, y'+\varepsilon f'+\alpha g')$

Differentiating J partially w. r. to $\in$ and put $\varepsilon=0$, we have

$$\left(\frac{\partial J}{\partial \varepsilon}\right)_{\varepsilon=0}=\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y} f+\frac{\partial G}{\partial y'} f'\right]dx$$

Put Your Own Notes

$$\int_{x_1}^{x_2}\frac{\partial G}{\partial y}fdx+\left[\frac{\partial F}{\partial y'}f\right]_{x_1}^{x_2}-\int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)fdx$$

$$\left(\frac{\partial J}{\partial \varepsilon}\right)_{\varepsilon=0}=\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]fdx \qquad \ldots(11)$$

Similarly $\left(\frac{\partial J}{\partial \alpha}\right)_{\alpha=0}=\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]gdx \qquad \ldots(12)$

Since $J(\varepsilon,\alpha)=k$

$$\frac{\partial J}{\partial \varepsilon}+\frac{\partial J}{\partial \alpha}.\frac{d\alpha}{d\varepsilon}=0$$

$$\left(\frac{d\alpha}{d\varepsilon}\right)_{\varepsilon=0}=-\frac{\left(\frac{\partial J}{\partial \varepsilon}\right)_{\varepsilon=0}}{\left(\frac{\partial J}{\partial \alpha}\right)_{\alpha=0}} \qquad \ldots(13)$$

While $\left(\frac{\partial J}{\partial \alpha}\right)_{\varepsilon=0}\neq 0$

By putting (11), (12), in (13), we get

$$\left(\frac{d\alpha}{d\varepsilon}\right)_{\varepsilon=0}=-\frac{\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]fdx}{\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]gdx} \qquad \ldots(14)$$

Now equation (8) & (14)

$$\Rightarrow -\frac{\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]fdx}{\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]gdx}=-\frac{\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]fdx}{\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]gdx}$$

$$\Rightarrow \frac{\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]fdx}{\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]fdx}=-\frac{\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]gdx}{\int_{x_1}^{x_2}\left[\frac{\partial G}{\partial y}-\frac{d}{dx}\left(\frac{\partial G}{\partial y'}\right)\right]gdx}=-\lambda(\text{say})$$

$$\int_{x_1}^{x_2}\left[\frac{\partial}{\partial y}(F\pm\lambda G)-\frac{d}{dx}\left\{\frac{\partial}{\partial y'}(F\pm\lambda G)\right\}\right]fdx=0 \qquad \ldots(15)$$

Put $H=F\pm\lambda G$

(15) reduces to

$$\int_{x_1}^{x_2}\left[\frac{\partial H}{\partial y}-\frac{d}{dx}\left(\frac{\partial H}{\partial y'}\right)\right]fdx=0$$

$$\Rightarrow \frac{\partial H}{\partial y}-\frac{d}{dx}\left(\frac{\partial H}{\partial y'}\right)=0$$

Example: Find the simple closed curve of a given parameter which encloses a maximum area.

(Dido's Problem)

Solution: Let l be the given length of the curve in the plane between two abscisssa

$a \& b$, then

$$\ell = \int_a^b \sqrt{1+y'^2}\,dx.$$

If A is the area between $x = a, b$ and $x -$ axis.

$$A = \int_a^b y\,dx \qquad \ldots(2)$$

Put $H = F + \lambda G$

$$H = y + \lambda\sqrt{1+y'^2}$$

$$\frac{\partial H}{\partial y} - \frac{d}{dx}\left(\frac{\partial H}{\partial y'}\right) = 0 \qquad \ldots(3)$$

$$1 - \frac{d}{dx}\left(\frac{\lambda y'}{\sqrt{1+y'^2}}\right) = 0$$

$$\frac{d}{dx}\left(\frac{\lambda y'}{\sqrt{1+y'^2}}\right) = 1$$

$$\frac{\lambda y'}{\sqrt{1+y'^2}} = x + c$$

$$\lambda y' = (x+c)\left(\sqrt{1+y'^2}\right)$$

$$\left[\lambda^2 - (x+c)^2\right] y'^2 = (x+c)^2$$

$$y' = \frac{x+c}{\sqrt{\lambda^2 - (x+c)^2}}$$

Put $\lambda^2 - (x+c)^2 = t$,

$$-2(x+c)\,dx = dt$$

$$\Rightarrow\ y = \frac{-1}{2}\int t^{-1/2}\,dt = -t^{1/2},$$

$$\Rightarrow\ y^2 = t\ \Rightarrow\ y^2 = \lambda^2 - (x+c)^2$$

$$\Rightarrow\ (x+c)^2 + y^2 = \lambda^2$$

The curve is a circle

Put Your Own Notes

Put Your Own Notes

Remark: $I = \int_{x_1}^{x_2} F(x, y, y')dx$

$J = \int_{x_1}^{x_2} G(x, y, y')dx = constant$

$\frac{\partial H}{\partial y} - \frac{d}{dx}\left(\frac{\partial H}{\partial y'}\right) = 0$, $H = F + \lambda G$

The problem of Extremizing I, when $J =$ constant (given) coincide with the problem of extremizing J under the constraint $I =$ constant such problems are known as reciprocal problems.

In general: If

$I = \int_{x_1}^{x_2} F(x, y_1, ------ y_n, y_1' --------, y_n')dx$

Subject to the conditions

$y_i(x_1) = a_i, y_i(x_2) = b_i,\ i = 1, 2, \ldots\ldots, n$ and the constants

$\int_{x_1}^{x_2} G(x, y_1, y_2 --- y_n, y_1' --------, y_n')dx = C_j$

$j = 1 -------- k < n$

Then the Euler's equation will be

$$\frac{\partial H}{\partial y} - \frac{d}{dx}\left(\frac{\partial H}{\partial y_i'}\right) = 0$$

$i = 1, 2, ------, n$

$H = F + \sum_{j=1}^{k} \lambda_j G_j \quad k = 1, 2 \ldots\ldots\ldots, n$

1.15. Variation Problems Involving More Independent Variables

Let us consider the double integral $I = \iint_R F(x, y, u, u_x, u_y)dxdy$ $\quad \ldots(1)$

taken over the region R in the $xy-$ plane

We have to find function $u(x, y)$, which makes (1) to be stationary under the prescribed condition on the curve C.

Let $u(x, y)$ be such a function. u take an arbitrary differentiable function $\eta = \eta(x, y)$ such that $\eta(x, y) = 0$ on C. (So that for any parameter ε, the function $u + \varepsilon\eta$ will take the same prescribed condition on C). Replacing u by $u + \varepsilon\eta$ in (1), we get

$$I(\varepsilon) = \iint_R F_\varepsilon dx\,dy \quad \ldots(2)$$

Where $F_\varepsilon = F(x, y, u + \varepsilon\eta, u_x + \varepsilon\eta_x, u_y + \varepsilon\eta_y)$ $\quad \ldots(3)$

$F = F(x, y, u, u_x, u_y)$

Put Your Own Notes

Then $I(\varepsilon)=I, F_\varepsilon=F$ at $\varepsilon=0$

Therefore I is stationary iff $I(\varepsilon)$ is stationary when $\varepsilon=0$

For (1) to be stationary, we have $\frac{dI(\varepsilon)}{d\varepsilon}=0$ we see that the integral in (2) independent of ε.

So $\frac{dI(\varepsilon)}{d\varepsilon}=\iint_R \frac{dF_\varepsilon}{d\varepsilon}dx.dy$ ... (4)

By (3)

$F_\varepsilon=F(x, y, u+\varepsilon\eta, u_x+\varepsilon\eta_x, u_y+\varepsilon\eta_y)$

Then $\frac{dF_\varepsilon}{d\varepsilon}=\frac{\partial F_\varepsilon}{\partial u}\eta+\frac{\partial F_\varepsilon}{\partial u_x}\eta_x+\frac{\partial F_\varepsilon}{\partial u_y}\eta_y$ ... (5)

$(4)\Rightarrow 0=\iint_R\left[\frac{\partial F_\varepsilon}{\partial u}\eta+\frac{\partial F_\varepsilon}{\partial u_x}\eta_x+\frac{\partial F_\varepsilon}{\partial u_y}\eta_y\right]dx\,dy$ ... (6)

Now $\iint_R \frac{\partial F_\varepsilon}{\partial u_x}\eta_x dxdy$

$=\iint_R \frac{\partial F_\varepsilon}{\partial u_x}\frac{\partial \eta}{\partial x}dx\ dy$

$=\int_{y=0}^{b}\left[\left(\frac{\partial F_\varepsilon}{\partial u_x}\eta(x,y)\right)_{x_1}^{x_2}-\int_{x_1(y)}^{x_2(y)}\frac{\partial}{\partial x}\left(\frac{\partial F_\varepsilon}{\partial u_x}\right)\eta(x,y))dx\right]dy$

$=-\iint_R \frac{\partial}{\partial x}\left(\frac{\partial F_\varepsilon}{\partial u_x}\right)\eta\ dxdy$

Similarly,

$\iint_R \frac{\partial F_\varepsilon}{\partial u_x}\eta_y dxdy=-\iint_R \frac{\partial}{\partial y}\left(\frac{\partial F_\varepsilon}{\partial u_y}\right)\eta\ dxdy$

Putting in (6) we get

$\iint_R\left[\frac{\partial F_\varepsilon}{\partial u}-\frac{\partial}{\partial x}\left(\frac{\partial F_\varepsilon}{\partial u_x}\right)-\frac{\partial}{\partial y}\left(\frac{\partial F_\varepsilon}{\partial u_y}\right)\right]\eta\, dxdy=0$

$\Rightarrow\left[\frac{\partial F_\varepsilon}{\partial u}-\frac{\partial}{\partial x}\left(\frac{\partial F_\varepsilon}{\partial u_x}\right)-\frac{\partial}{\partial y}\left(\frac{\partial F_\varepsilon}{\partial u_y}\right)\right]=0$

at $\varepsilon=0\Rightarrow\frac{\partial F}{\partial u}-\frac{\partial}{\partial x}\left(\frac{\partial F}{\partial u_x}\right)-\frac{\partial}{\partial y}\left(\frac{\partial F}{\partial u_y}\right)=0$ (ostrogradaskey equation)

Put Your Own Notes

Example: Write the ostrogradaskey equation for the functional

(a) $$I\left[z(x,y)\right]=\iint_D\left[\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2\right]dx\,dy$$

(b) $$I\left[z(x,y)\right]=\iint_D\left[\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2+2zf(x,y)\right]dx\,dy$$

(c) $$I\left[z(x,y)\right]=\iint_D\left[\left(\frac{\partial z}{\partial x}\right)^2-\left(\frac{\partial z}{\partial y}\right)^2\right]dx\,dy$$

1.16. Variational Problems with Moving Boundaries

We have three different cases.

Case I: Suppose that the end points (x_0, y_0) and (x_1, y_1) are moving on two vertical lines $x = x_0$ and $x = x_1$ respectively.

We have for a functional

$$J=\int_{x_0}^{x_1}F(x,y,y')dx$$

The general variation in J

$$\delta J=\int_{x_0}^{x_1}h\left[\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]dx+\left[\left(F-y'\frac{\partial F}{\partial y'}\right)\delta x\right]_{x_0}^{x_1}+\left[\frac{\partial F}{\partial y'}\delta y\right]_{x_0}^{x_1} \qquad \ldots(1)$$

Since the basic condition for an extremum of J with fixed end points is

$$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0 \qquad \ldots(2)$$

The condition $\delta J = 0$, by (1)

$$\left[\left(F-y'\frac{\partial F}{\partial y'}\right)\delta x\right]_{x_0}^{x_1}+\left[\frac{\partial F}{\partial y'}\delta y\right]_{x_0}^{x_1}=0 \qquad \ldots(3)$$

Now, $\delta x_0 = 0,\ \delta x_1 = 0$

($\because$ The end points moves along the vertical line)

So (3) reduces to

$$\left[\frac{\partial F}{\partial y'}\delta y\right]_{x_0}^{x_1}=0 \qquad \ldots(4)$$

Since $\delta y = h(x)$, (4) becomes

$$\left(\frac{\partial F}{\partial y'}\right)_{x_1}h(x_1)-\left(\frac{\partial F}{\partial y'}\right)_{x_0}h(x_0)=0 \qquad \ldots(5)$$

(5) must be true for all $h(x)$, where $h(x)$ is arbitrary, therefore we get

Put Your Own Notes

$$\left(\frac{\partial F}{\partial y'}\right)_{x=x_1}=0 \quad \& \quad \left(\frac{\partial F}{\partial y'}\right)_{x=x_0}=0 \qquad \ldots(6)$$

Therefore in this case J to be extremum, we have to find the solution of the Euler equation and the two constant in the solution of Euler equation: obtained by using contains (6)

(6) are called natural boundary conditions

Case II: Let (x_0, y_0) be fixed and (x_1, y_1) move along a vertical line $x = x_1$.

Thus we obtain as in **Case-I**, the Euler equation must be satisfied, and the condition $\delta J = 0$ becomes $\left[\left(F - y'\frac{\partial F}{\partial y'}\right)\delta x\right]_{x_0}^{x_1} + \left[\frac{\partial F}{\partial y'}\delta y\right]_{x_0}^{x_1} = 0$

Since $\delta x_0 = 0 = \delta x_1$, also $(\delta y)_{x_0} = 0$

Then we get

$$\left[\frac{\partial F}{\partial y'}\delta y\right]_{x=x_1} = 0 \text{ this holds for all } \delta y$$

So we get

$$\left(\frac{\partial F}{\partial y'}\right)_{x=x_1}=0$$

In this case the two constants can be obtained by

(i) When $x = x_0,\ y = y_0$ (one end is fixed)

(ii) $\left(\frac{\partial F}{\partial y'}\right)_{x=x_1}=0$

Case III: Suppose that (x_0, y_0) is fixed and (x_1, y_1) is moving along certain curve $y = f(x)$

As before, we have the basic conditions

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$$

The condition $\delta J = 0$, gives

$$\left[\left(F - y'\frac{\partial F}{\partial y'}\right)\delta x\right]_{x_0}^{x_1} + \left[\frac{\partial F}{\partial y'}\delta y\right]_{x_0}^{x_1} = 0$$

Since (x_0, y_0) is fixed, we have

$$(\delta x)_{x_0} = 0 = (\delta y)_{x_0}$$

Put Your Own Notes

We have

$$\left[\left(F - y'\frac{\partial F}{\partial y'}\right)\delta x\right]_{x=x_1} + \left[\frac{\partial y}{\partial y'}\delta y\right]_{x=x_1} = 0$$

$$\left[\left(F - y'\frac{\partial F}{\partial y'}\right)\delta x\right]_{x=x_1} + \left[\frac{\partial y}{\partial y'}f'(x)\delta y\right]_{x=x_1} = 0$$

Since $y = f(x)$

$\Rightarrow \delta y = f'(x)\,\delta x$

$$\left[F + \{f'(x) - y'\}\frac{\partial F}{\partial y'}\right]_{x=x_1}\delta x_1 = 0 \qquad \ldots(*)$$

Since δx_1 varies arbitrary, we get

From $(*)$ $\quad F + (f' - y')\dfrac{\partial F}{\partial y'} = 0 \quad \rightarrow$ Transversality condition, one of the constants in the solution of Euler's equation can be obtained by the condition $y(x_0) = y_0$.

The other constant is obtained by eliminating x_1 between the equation $y = f(x)$ and $(*)$.

And similarly, if (x_0, y_0) is moving along $y = \psi(x)$, we can find the transversality condition

$$\left[F - \left((\psi' - y')\frac{\partial F}{\partial y'}\right)\right] = 0$$

Which must be satisfied by the point (x_0, y_0).

Example: Find the curve on which on extremum of the functional

$$I = \int_0^{\pi/4}\left(y^2 - y'^2\right)dx$$

$y(0) = 0$

can be achieved if the second boundary points is permitted to move along the straight line $x = \dfrac{\pi}{4}$.

Solution: $F = y^2 - y'^2$

Euler equation is

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$$

$$2y - \frac{d}{dx}(-2y') = 0$$

$$y'' + y = 0$$

Put Your Own Notes

$y = c_1 \cos x + c_2 \sin x$ ...(1)

At $x = 0,\ y = 0,\ 0 = c_1 \cos 0 + c_2 \sin 0$

$\Rightarrow\ c_1 = 0$

(i) reduces to

$y = c_2 \sin x$ ...(2)

The natural boundary condition $\left(\frac{\partial F}{\partial y'}\right)_{x=\frac{\pi}{4}} = 0$

$(-2c_2 \cos x)_{x=\frac{\pi}{4}} = 0 \Rightarrow\ c_2 = 0$

final equation of the required curve is $y = 0$

Example 2: Find the transversality condition for

$v = \int_{x_0}^{x_1} A(x, y)\sqrt{1+y'^2}\,dx$

Solution: We know that the transversality condition is

$F - (\varphi' - y')\frac{\partial F}{\partial y'} = 0$

Where $F = A\sqrt{1+y'^2}$

We get

$A\sqrt{1+y'^2} - (\phi' - y')\frac{Ay'}{\sqrt{1+y'^2}} = 0$

$A(1+y'^2) + (\phi' - y')Ay' = 0$

$A(1\ + y'^2 + \phi' y' - y'^2)\ =$

$\Rightarrow\ 1 + \phi' y' = 0 \Rightarrow y' = -\frac{1}{\phi'}$

Which is the condition of orthogonality.

In general, when F is of the form $F = A\sqrt{1+y'^2}$ then the transvarsality condition reduced to orthogonality condition

Exercise:

(i) Test for an extremum of the functional

$$I = \int_0^{x_1} \frac{\sqrt{1+y'^2}}{y}dx$$

Given that $y(0) = 0$ and $y_1 = x_1 - 5$

(ii) Find the curve on which an extremum

$$v[y(x)] = \int_0^{x_1} \frac{\sqrt{1+y'^2}}{y}dx,$$

$y(0)=0$,

Can be achieved if the second boundary ference of the circle $(x-9)^2+y^2=9$

Put Your Own Notes

1.17. Calculus of the Variation (Problems)

Find the extremals for the following

(i) $I=\int_{x_1}^{x_2}\left(2xy+y'''^2\right)dx$

(ii) $I=\int_0^1\left(1+y''^2\right)dx$, $\quad y(0)=0,\ y'(0)=1,\ y(1)=1,\ y'(1)=1$.

(iii) $I=\int_0^{h/2}\left(y''^2-y^2+x^2\right)dx$, $\quad y(0)=1, y'(0)=0, y(h/2)=0, y'(h/2)=1$

(iv) $I=\int_{x_0}^{x_1}\left(16y^2-y''^2+x^2\right)dx$

(v) Find the curve joining points (0, 0) and (1, 0), for such the integral $\int_0^1 y''^2 dx$ is a minimum if $y'(0)=a,\ y'(1)=b$.

(vi) $I=\int_{x_1}^{x} y'(1+x^2y')dx$

(vii) $I=\int_{x_0}^{x_1} y'(x+y')\,dx$

(viii) $I=\int_{x_0}^{x_1}\left[y^2+y'^2-2y\sin x\right]dy$

(ix) On what curves; the functional $I=\int_0^1\left(y'^2+12xy\right)dx$ with $y(0)=0$ and $y(1)=1$ be extremized.

(x) $I=\int_{x_0}^{x_1}\left(y^2-y'^2-2y\sin x\right)dx$

(xi) If $=\int_1^2\frac{x^2}{y'^2}dx$, $y(1)=1,\ y(2)=4$

(xii) $I=\int_{x_0}^{x_1}\frac{y'^2}{x}dx$

(xiii) $I=\int_{x_0}^{x_1}\left(y'+2yy'-16y^2\right)dx$

(xiv) $I=\int_{x_0}^{x_1}\frac{\sqrt{1+y'^2}}{y}dx$

(xv) $I=\int_{x_0}^{x_1}\left(y^2+2xyy'\right)dx$, $\ y(x_0)=y_0,\ \ y(x_1)=y_1$

(xvi) $I=\int_0^1\left(xy+y^2y'\right)dx \qquad y(0)=1,\ \ y(1)=2$

Put Your Own Notes

(xvii) $I=\int_{x_1}^{x_2}\frac{y-xy'}{y^2}dx,\quad y(x_1)=y_1,\ y(x_2)=y$

(xviii) $I=\int_{x_1}^{x_2}\frac{2yx-y^2y'}{x^2}dx,\qquad y(x_1)=y_1\ \&\ y(x_2)=y$

(xix) $I=\int_0^{h/2}\left(y'^2-y^2\right)dx,\qquad y(0)=0,\ y(h/2)=1$

(xx) $I=\int_0^1\left(x'^2+1\right)dt,\qquad x(0)=1,\ x(1)=2$

(xxi) $I=\int_1^2\frac{\sqrt{1+y'^2}}{x}dx,\qquad y(1)=0,\ y(2)=1$

(xxii) $I=\int_{x_2}^{x_1}\left(y^2+y'^2+2y\,e^x\right)dx$

(xxiii) $I=\int_{x_1}^{x_2}\frac{\sqrt{1+y'^2}}{x}dx$

(xxiv) $I=\int_{x_0}^{x_1}(y+xy')dx$

(xxv) $I=\int_{x_0}^{x_1}\left(x^2y'^2+2y^2+2xy\right)dx.$

(xxvi) $I=\int_0^{h/2}\left[\left(\frac{dy}{dt}\right)^2-y^2+2+y\right]dt,\quad y(0)=0,\ y(h/2)=0$

(xxvii) $I=\int_0^1\left(y^2+x^2y'\right)dx,\quad y(0)=0,\ y(1)=0$

(xxviii) $I=\int_{x_1}^{x_2}\left(xy^2+x^2yy'\right)dx,\quad y(x_1)=y_1,\ y(x_2)=y$

Find the extremals of the isopeimetric problems,

(i) $I=\int_0^1\left(y'^2+x^2\right)dx$,

given that

$\int_0^1 y^2dx=2$

$y(0)=0,\ y(1)=1$

(ii) $I=\int_{x_0}^{x_1}y'^2\,dx$, given that

$\int_{x_0}^{x_1}y\,dx=a(constt.)$

(iii) $I=\int_0^h\left(y'^2-y^2\right)dx$ if $\int_0^h y\,dx=1$

$y(0)=0,y(h)=1$

Put Your Own Notes

(iv) Find the extremals of the functional

$$I=\int_0^1\left(y'^2+z'^2-4xz'-4z\right)dx$$

subject to the condition $y(0)=0,\ y(1)=1,\ z(0)=0,\ z(1)=1$ and the constraints

$$\int_0^1\left(y'^2-xy'-z'^2\right)dx=2$$

(v) Minimize the functional $L=\frac{1}{2}\int_0^2 (x)^2\,dt$, where $x=x(t)$

$x(0)=1,\ x(0)=1,\ x(2)=2,\ x(2)=0$

(vi) Find the curve on which an extremum of the functional

$$I=\int_0^{h/4}\left(y^2-y'^2\right)dx,\ y(0)=0.$$

can be achieved, if the second boundary point is permitted to move along the straight line $x=\frac{\pi}{4}$.

(vii) Test for an extremum the functional

$$I=\int_0^{x_1}\frac{\sqrt{1+y'^2}}{y}dx$$

Given that $y(0)=0$ and $y_1=x_1-5$

1.18. Variational Methods of Solving Ordinary Differential Equations (Rayleigh-Ritz Method)

Suppose, we have the boundary value problem

$$\alpha_0(x)y''+\alpha_1(x)y'+\alpha_2(x)y=f(x) \quad \ldots(1)$$

with the boundary conditions

$$y(a)=y_a \text{ and } y(b)=y_b \quad \ldots(2)$$

We construct the function $F(x,y,y')$ such that equation (1) is the Euler's equation for the extrema of the functional.

$$I=\int_a^b F(x,y,y')dx \quad \ldots(3)$$

With the choice of suitable functions $y_0(x),\ y_1(x),\ y_2(x),\ldots$, we get.

$$y(x)=y_0(x)+c_1y_1(x)+c_2y_2(x)+\ldots\ldots+c_ny_n(x). \quad \ldots(4)$$

The function $y_0(x)$ is the simplest function that satisfies the given boundary conditions (2), whereas $y_1(x),\ y_2(x)\ldots\ldots$ are linearly independent functions satisfying the homogeneous boundary conditions.

$$y(a)=0,\ y(b)=0 \quad \ldots(5)$$

Put Your Own Notes

Substituting (4) in (3) and carrying out the integration, we find $I = I(c_1, c_2, .., c_n)$ a function of the n constants $c_1, c_2, \ldots, c_n$. These constants are chosen so that I is extremum. The necessary conditions are

$$\frac{\partial I}{\partial c_1} = 0, \frac{\partial I}{\partial c_2} = 0, \ldots, \frac{\partial I}{\partial c_n} = 0 \qquad \ldots (6)$$

Solving these n simultaneous equations we find $c_1, c_2, \ldots, c_n$ and hence obtain the approximate solution. This method is known as Rayleigh-Ritz method(*****).

1.19. Remark (Solution of Differential Equation)

Consider,

$$a(x)y'' + b(x)y' + c(x)y = f(x) \qquad \ldots (1)$$

Multiplying both sides by an arbitrary function $\mu(x)$, such that

$$-2p = a\mu,\ 2p' = b\mu,\ 2Q = c\mu,\ -R = f\mu \qquad \ldots (2)$$

$$\Rightarrow\ p' - \frac{b}{a}p = 0 \Rightarrow P = \exp\int_{x_0}^{x_1} \frac{b}{a}dx \quad (\text{if } P = 1 \text{ at } x_0)$$

$$\text{and } \mu = \frac{-2}{b}p' \ \Rightarrow\ \mu = -\frac{2}{a}p$$

$$\&\ Q = \frac{-c}{a}p, R = \frac{2f}{a}p \quad p = p(x)$$

It can be seen that the required functional is of the form

$$I[y] = \int\left[P(x)y'^2 + Q(x)y^2 + R(x)y\right]dx \qquad \ldots (3)$$

with appropriate limits

Now we can solve the integral by Rayleigh – Ritz method.

Example 1: Find the approximate solution of $y'' = 1,\ y(0) = y(1) = 0$

The given equation is the Euler's equation of the variational problem of optimizing the functional.

$$I = \int_0^1 \left(y'^2 + 2y\right)dx$$

We take the solution as

$$y(x) = \left(x - x^2\right)\left(c_1 + c_2 x + c_3 x^2 + \ldots + c_n x^n\right)$$

With one –term approximation, we have

$$y(x) = c\left(x - x^2\right)$$

$$y'(x) = c(1 - 2x)$$

Substituting these expressions in I, we get

$$I = \int_0^1 \left[c^2(1-2x)^2 + 2c\left(x - x^2\right)\right]dx$$

$$\frac{dI}{dc}=0 \Rightarrow \int_0^1\left[2c(1-2x)^2+2(x-x^2)\right]dx=0$$

$$\Rightarrow c=-\frac{1}{2}$$

Therefore the desired solution is $y(x)=\frac{x^2-x}{2}$

Identically, we note that this is the exact solution.

Example 2: Solve $y''+y=-x,\ \ y(0)=y(1)=0$

The variational problem is optimizing the functional.

$$I=\int_0^1\left(y'^2-y^2-2xy\right)dx$$

Taking $y=c\left(x-x^2\right)$ and $y'=c(1-2x)$,

$$I=\int_0^1\left[c^2(1-2x)^2-c^2\left(x-x^2\right)^2-2cx\left(x-x^2\right)\right]dx$$

$$=\int_0^1\left[c^2(1-4x+4x^2-x^2+2x^3-x^4)-2c\left(x^2-x^3\right)\right]dx$$

$$=\frac{3}{10}c^2-\frac{c}{6}$$

$$\frac{dI}{dc}=0 \Rightarrow C=\frac{5}{18}$$

The approximate solution is $y(x)=\frac{5}{18}\left(x-x^2\right)$

This solution yields $y(0.5)=0.069$ whereas the exact solution $y(x)=\frac{\sin x}{\sin 1}-x$ yields $y(0.5)=0.070$.

Example 3: Solve $y''+xy=-x,\ \ y(0)=y(1)=0$

The corresponding variation problem is to optimize $I=\int_0^1\left(y'^2-xy^2-2xy\right)dx$ choosing $y(x)=c\left(x-x^2\right),\ y'(x)=c(1-\ 2x)$, we get

$$I=\int_0^1\left[c^2(1-2x)^2-c^2x(x-x^2)^2-2cx(x-x^2)\right]dx$$

$$\frac{dI}{dc}=0 \Rightarrow \int_0^1\left[c\left\{(1-2x)^2-x\left(x-x^2\right)^2\right\}-x\left(x-x^2\right)\right]dx=0$$

$$\therefore\ 19c=5 \text{ or } c=\frac{5}{19}$$

The approximate solution is $y(x)=\frac{5}{19}\left(x-x^2\right)$

Starting with better approximation.

$$y(x)=\left(x-x^2\right)(c_1+c_2x)$$

Put Your Own Notes

Put Your Own Notes

We find $c_1 = 0.177,\ c_2 = 0.173$ and the approximation becomes

$$y(x) = (0.177 + 0.173x)x(1-x)$$

Example 4: Solve $xy'' + y' + y = x,\ y(0) = 0,\ y(1) = 1$

Here the variational problem is to optimize the functional

$$I = \int_0^1 \left(xy'^2 - y^2 + 2xy\right)dx$$

Choosing $y(x) = x + c\left(x - x^2\right)$, we satisfy the given boundary conditions.

With $y = (1+c)x - cx^2$ and $y' = 1 + c - 2cx$, we find

$$I = \int_0^1 \left[(1+c)^2 x + 4c^2x^3 - 4c(1-c)x^2 - (1+c)^2x^2 - c^2x^4 + 2c(1+c)x^3 + 2(1+c)x^2 - 2cx^3\right]dx$$

$$\frac{dI}{dc} = 0 \Rightarrow c = \frac{5}{4}$$

$\therefore$ The approximate solution is

$$y(x) = x + \frac{5}{4}(x - x^2) = (9 - 5x)\frac{x}{4}$$

A better approximation can be obtained by choosing $y(x)$ as

$$y(x) = x + \left(x - x^2\right)\left(c_1 + c_2 x\right)$$

We find $c_1 = \dfrac{85}{26},\ c_2 = -\dfrac{35}{13}$ giving

$$y(x) = \frac{x(111 - 155x + 70x^2)}{26}$$

Example 5: Solve $y'' + \left(1 + x^2\right)y + 1 = 0,\ y(\pm 1) = 0$

The corresponding variational problem is to optimize the functional

$$I = \int_{-1}^1 \left[y'^2 - \left(1 + x^2\right)y^2 - 2y\right]dx$$

By observing the differential equation and the boundary conditions, we note that the solution has to be an even function of x.

Choosing $y(x) = c\left(1 - x^2\right)$ and $y' = c(1 - 2x)$ we find

$$I = \int_{-1}^1 \left[c^2(1-2x)^2 - \left(1+x^2\right)c^2\left(1-x^2\right)^2 - 2c\left(1-x^2\right)\right]dx$$

Evaluating the integral, we have

$$I = 8\left(\frac{19}{105}c^2 - \frac{c}{3}\right)$$

$$\frac{dI}{dc} = 0 \Rightarrow c = \frac{35}{38}$$

Put Your Own Notes

i.e., $y(x)=\frac{35}{38}\left(1-x^2\right)$

is the one term approximation.

Taking the two –term approximation as

$$y(x)=\left(1-x^2\right)\left(c_1+c_2x^2\right)$$

We find

$$I=8\left[\frac{19}{105}c_1^2+\frac{78}{945}c_1c_2+\frac{331}{3465}c_2^2-\frac{c_1}{3}-\frac{c_2}{15}\right]$$

Now, $\frac{\partial I}{\partial c_1}=0, \frac{\partial I}{\partial c_2}=0$ yield

$$\frac{38}{105}c_1+\frac{78}{945}c_2=\frac{1}{3}$$

$$\frac{78c_1}{945}+\frac{662}{3465}c_2=\frac{1}{5}$$

Solving we get $c_1=0.933,\ c_2=-0.054$

$$y(x)=\left(1-x^2\right)\left(0.993-0.054x^2\right)$$

Note: Ralyleigh- Ritz method can also be used to find approximately the smallest Eigen value.

Example 7: Find approximately the smallest eigen value λ of

$$y''+\lambda y=0,\ \ y(0)=y(1)=0$$

Here

$$I=\int_0^1\left(y'^2-\lambda y^2\right)dx$$

Choosing $y(x)=c\left(x-x^2\right)$ and $y'(x)=c(1-2x)$ we find

$$I=\int_0^1\left[c^2(1-2x)^2-\lambda c^2(x-x^2)^2\right]dx$$

$$\frac{dI}{dc}=0\Rightarrow\lambda=\frac{\int_0^1\left(1-4x+4x^2\right)dx}{\int_0^1\left(x^2-2x^3+x^4\right)dx}=10$$

The exact eigen values are $\lambda_n=n^2\pi^2$ and least eigen value is $\lambda_1=\pi^2=9.870$

Example 8: Solve $\left(1-x^2\right)y''-2xy'+2y=0,\ y(0)=0,\ y(1)=1$

Here $I=\int_0^1\left[\left(1-x^2\right)y'^2-2y^2\right]dx$

Taking $y(x)=x+c\left(x-x^2\right)$ and $y'(x)=1+c(1-2x)$ we find

Put Your Own Notes

$$I=\int_0^1\left[\left(1-x^2\right)\left\{(1+c)^2-4c(1+c)x+4c^2x^2\right\}-2\left\{(1+c)^2x^2-2c(1+c)x^3+c^2x^4\right\}\right]dx$$

$$\frac{dI}{dc}=0\Rightarrow$$

$$0=\int_0^1\left[\left(1-x^2\right)\left\{2(1+c)-4(1+2c)x+8cx^2\right\}-2\left\{2(1+c)x^2-2(1+2c)x^3+2cx^4\right\}\right]dx$$

Integrating, we find $c=0$

$\therefore$ The desired solution is $y(x)=x$, which is exact.

Example 9: Find the smallest eigen value of $y''+\lambda y=0,\ y(\pm 1)=0$

Here $I=\int_{-1}^{1}\left(y'^2-\lambda y^2\right)dx$

Choosing $y=c\left(1-x^2\right)$ and $y'(x)=c(-2x)$ we find

$$I=\int_{-1}^{1}\left[4c^2x^2-\lambda\left(1-2x^2+x^4\right)c^2\right]dx$$

$$\frac{dI}{dc}=0\Rightarrow\lambda=\frac{\int_{-1}^{1}4x^2dx}{\int_{-1}^{1}\left(1-2x^2+x^4\right)dx}\Rightarrow\lambda=\frac{15}{2}$$

The exact value is $\frac{\pi^2}{4}=2.467$

Example 10: Find the smallest Eigen value of

$(1+x)y''+y'+\lambda y=0,\ y(0)=y(1)=0$

Here $I=\int_0^1\left[(1+x)y'^2-\lambda y^2\right]dx$

Choosing $y(x)=c\left(x-x^2\right)$ and $y'(x)=c(1-2x)$ we find

$$I=c^2\int_0^1\left[(1+x)(1-2x)^2-\lambda\left(x-x^2\right)^2\right]dx$$

$$\frac{dI}{dc}=0\Rightarrow\lambda=\frac{\int_0^1(1+x)(1-2x)^2\,dx}{\int_0^1\left(x-x^2\right)^2dx}$$

$$\frac{dI}{dc}=0\Rightarrow\lambda=\frac{\int_0^1(1+x)(1-2x)^2dx}{\int_0^1(x-x^2)dx}$$

Taking $y(x)=\left(x-x^2\right)\left(c_1+c_2x\right)$, we obtain $\lambda_1=14.42,\lambda_2=63.6$ as the smallest two eigen values.

Example 11: Solve $y''+y=-\frac{x^3}{5},\ y(0)=0,\ y(1)=1$

Here $I=\int_0^1\left(y'^2-y^2-\frac{2}{5}x^3y\right)dx$

Put Your Own Notes

Choosing $y(x)=x+c\left(x-x^2\right)$, we get

$$I=\int_0^1\left[(1+c-2cx)^2-\left\{(1+c)x-cx^2\right\}^2-\frac{2}{5}x^3\left\{(1+c)x-cx^2\right\}\right]dx$$

$$\frac{dI}{dc}=0\Rightarrow I=\int_{-1}^{1}\int_{-1}^{1}\left[4c^2x^2\left(1-2y^2+y^4\right)+4c^2y^2\left(1-2x^2+x^4\right)\right.$$

$\therefore\ 10c-3=0$, by evaluating the integral or $c=\dfrac{3}{10}$.

The solution is $y(x)=\dfrac{13x-3x^2}{10}$. This yields $y(0.5)=\dfrac{23}{40}$. This value agrees with that of the value obtained from the exact solution $y(x)=\dfrac{6x-x^3}{5}$.

In the problems considered so far , the boundary conditions deal with prescribed values of the unknown function $y(x)$, like $y(a)=y_{-a},\ y(b)=y_b$

In the derivative of the unknown function enters the boundary conditions, then Rayleigh- Ritz method needs modification. This will be discussed in the next section.

Galerbin Method: (Conversion of ODE into variational problem to find approximate solution).

Galerbin method convert the ODE into functional (variational problem) and then using Rayleigh Ritz method to find the approximate solution.

Method: Consider the second order differential equation

$$a(x)y''+b(x)y'+c(x)y=f(x)$$

Corresponding functional of this differential equation is given by

$$I=\int_{x_1}^{x_2}\left[P(x)y'^2+Q(x)y^2+R(x)y\right]dx$$

Where $P=\int\dfrac{b}{a}dx$

$$Q=-\frac{cP}{a}$$

$$R=\frac{2fP}{a}$$

Example: Find the approximate solution of this differential equation $y''=1$ at boundary condition $y(0)=0,\ y(1)=0$.

Solution. : Consider a second order differential equation

$$a(x)y''+b(x)y'+c(x)y=f(x) \quad \ldots(1)$$

$$y''=1 \quad \ldots(2)$$

Put Your Own Notes

Compare equation (1) & (2)

We get $a=1,\ b=0,\ c=0,\ f=1,$

$P=e^0=1$

$Q=0$

$R=2$

In equation (1)

$$I=\int_{x_1}^{x_2}\left[P(x)y'^2+Q(x)y^2+R(x)y\right]dx \qquad \ldots(3)$$

Putting the value in equation (3)

$$I=\int_0^1\left(y'^2+2y\right)dx \quad y(0)=0,\ y(1)=0 \qquad \ldots(4)$$

Where $y=cx(1-x)$ $\qquad \ldots(*)$

$y'=c(1-2x)$

In equation (4)

$$=\int_0^1\left[c^2(1-2x)^2+2cx(1-x)\right]dx$$

$$=\int_0^1\left[c^2-4c^2x+4x^2c^2+2cx-2cx^2\right]dx$$

$$=\left[c^2x-4c^2\frac{x^2}{2}+4\frac{x^3}{3}c^2+2c\frac{x^2}{2}-2c\frac{x^3}{3}\right]_0^1$$

$$=c^2-4\frac{c^2}{2}+4\frac{c^2}{3}+c-2\frac{c}{3}$$

$$I=c^2\left(\frac{1}{3}\right)+\frac{c}{3} \qquad \ldots(5)$$

Differentiate equation (5) with respect to c

$$\frac{dI}{dc}=0 \Rightarrow \frac{2c}{3}+\frac{1}{3}=0$$

$$c=-\frac{1}{2}$$

Putting the value in $(*)$, $y=-\frac{1}{2}x(1-x)$

Direct Substitution Method

$y''\to -y'^2$

$y\to y^2.$

Put Your Own Notes

$y' \to 0$

x or constant $\to$ multiply by $2y$.

Example: Find the smallest Eigen value approximately using Rayleigh Ritz method.

$y'' + \lambda y = 0$

$y(0) = y(1) = 0$

Solution: Direct Substitution method:-

$y'' \to -y'^2$

$y \to y^2$

$$-y'^2 + \lambda y^2 = 0 \quad \ldots(1)$$

$$I = \int_0^1 \left(y'^2 - \lambda y^2\right) dx \quad \ldots(2)$$

Put $y = cx(1-x)$

$y' = c(1-2x)$

$$I = \int_0^1 \left[c^2(1-2x)^2 - \lambda c^2 x^2 (1-x)^2\right] dx$$

$$I = c^2 \int_0^1 \left[(1-2x)^2 - \lambda x^2 (1-x)^2\right] dx$$

$$\frac{dI}{dc} = 0$$

$$0 = 2c\left[\int_0^1 (1-2x)^2 - \lambda x^2 (1-x)^2\right] dx$$

$$\lambda = \frac{\int_0^1 (1-2x)^2\, dx}{\int_0^1 x^2 (1-x)^2\, dx}$$

$$\lambda = \frac{\left[x - 4\frac{x^2}{2} + 4\frac{x^3}{3}\right]_0^1}{\left[\frac{x^3}{3} - 2\frac{x^4}{4} + \frac{x^5}{5}\right]_0^1}$$

$$\lambda = \left[\frac{1}{3}\right] \times \left[\frac{30}{10-15+6}\right]$$

$\lambda = 10$.

Put Your Own Notes

1.20. Variational Methods of Solving Partial Differential Equations (Rayleigh-Ritz Method)

Consider the boundary value problem $-(pu_x)_x - (pu_y)_y + qu = r$ in the region R bounded by the closed curve C. Let $u(x)$ be prescribed on C. The corresponding the boundary variational problem is to optimize the functional.

$$\iint_R \left[p(u_x^2 + u_y^2) + qu^2 - 2ru \right] dxdy$$

We take the trial function such that it satisfies the given boundary condition.

Example 2: Solve Laplace equation in a square $u_{xx} + u_{yy} = 0$ $|x| \le 1,\ |y| \le 1$

$u = 0$ on $|x| = 1$, $u = 1 - x^2$ on $|y| = 1$

Trial solution in this case is $u = (1 - x^2)\left[1 + c(1 - y^2)\right]$

The required functional is $I = \int_{-1}^{1}\int_{-1}^{1}\left(u_x^2 + u_y^2\right)dx$

$$= \int_{-1}^{1}\int_{-1}^{1}\left[4c^2y^2(1 - 2x^2 + x^4)\right] + 4x^2\left[1 + 2c(1 - y^2) + c^2(1 - 2y^2 + y^4)\right]dx\,dy$$

$$\frac{dI}{dc} = 0$$

$$\Rightarrow\ 0 = \int_{-1}^{1}\int_{-1}^{1}\left[8cy^2(1 - 2x^2 + x^4) + 4x^2\left\{2(1 - y^2) + 2c(1 - 2y^2 + y^2)\right\}\right]dx\,dy$$

Evaluating the integral, we find $c = -\frac{5}{8}$

and $u(x, y) = 1 - x^2 - \frac{5}{8}\left(1 - x^2\right)\left(1 - y^2\right)$ as the desired solution.

We also get $u(0,\ 0) = \frac{3}{8} = 0.375$

Whereas the exact value is 0.411.

Example 3: Find the smallest eigen value of $u_{xx} + u_{yy} + \lambda u = 0$, $|x| \le 1$, $|y| \le 1$

Hence $J = \int_{-1}^{1}\int_{-1}^{1}(u_x^2 + u_y^2 - \lambda u^2)dx\,dy$

Taking $u(x, y) = c\left(1 - x^2\right)\left(1 - y^2\right)$,

$$J = \int_{-1}^{1}\int_{-1}^{1}\left[4x^2c^2(1 - y^2)^2 + 4y^2c^2(1 - x^2) - \lambda c^2(1 - x^2)^2(1 - y^2)^2\right]dx\,dy$$

$$= \left(\frac{256}{45} - \frac{256\lambda}{225}\right)c^2$$

$$\frac{dJ}{dc} = 0 \Rightarrow \lambda = 5$$

$\therefore\ \lambda = 5$ is the least eigen value

Put Your Own Notes

The exact value $=\dfrac{\pi^2}{2}=4.93$

Example 4: Solve Poisson's equation in a circle $u_{xx}+u_{yy}=-1,\ x^2+y^2\le 1$

$u=0$ on $x^2+y^2=1$

Here $J=\iint\limits_R (u_x^2+u_y^2-2u)\,dx\,dy$

Where R is the region $x^2+y^2\le 1$

Choosing $u(x,y)=c\left(1-x^2-y^2\right)$, we find

$$J=\iint\limits_R\left[4c^2(x^2+y^2)-2c(1-x^2-y^2)\right]dx\,dy$$

$$\frac{J}{2\pi}=\int_0^1 rc^2r^2r\,dr-\int_0^1 2c(1-r^2)r\,dr=c^2-\frac{c}{2}$$

$$\frac{dJ}{dc}=0\Rightarrow c=\frac{1}{4}$$

$u(x,y)=\dfrac{1-x^2-y^2}{4}$, exact solution.

Example 5: Solve: Poisson's equation in an ellipse $u_{xx}+u_{yy}=-1$, $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}\le 1$ $u=0$ on $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$

Here $J=\iint\limits_R (u_x^2+u_y^2-2u)\,dx\,dy$

Where R is the region insider $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$

Taking $u(x,y)=c\left(1-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}\right)$,

we find $J=\iint\limits_R 4c^2\left(\dfrac{x^2}{a^4}+\dfrac{y^2}{b^4}\right)dx\,dy-2c\iint\limits_R\left(1-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}\right)dx\,dy$

$$\frac{dJ}{dc}=0\Rightarrow 4c=\frac{\iint\limits_R\left(1-\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}\right)dx\,dy}{\iint\limits_R\left(\dfrac{x^2}{a^4}+\dfrac{y^2}{b^4}\right)dx\,dy}$$

Changing the coordinates by the transformation $x=ar\cos\theta,\ y=br\sin\theta$,

we find $4c=\dfrac{\int_0^1(1-r^2)r\,dr}{\int_0^1 r^3\,dr}\times\dfrac{2}{\dfrac{1}{a^2}+\dfrac{1}{b^2}}$

Put Your Own Notes

$$\therefore \quad c=\frac{a^2b^2}{2(a^2+b^2)}$$

$$u(x,y)=\frac{a^2b^2}{2(a^2+b^2)}\left(1-\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)$$

Incidentally this solution is the exact solution.

Example 6: Poisson's equation in an equilateral region: Solve $u_{xx}+u_{yy}=1$

Inside the region R of the equilateral triangle bounded by the lines $y=\pm\frac{x}{\sqrt{3}}$ and $x=a$, given that u vanishes on the boundary.

Here $J=\iint_R (u_x^2+u_y^2+2u)dxdy$

Choosing $u(x,y)=c\left(x^2-3y^2\right)(a-x)$, we get

$$J=c^2\iint_R\left[(2ax-3x^2+3y^2)^2+36y^2(x-a)^2\right]dx\,dy$$

$$-2c\iint_R(ax^2-x^3-3ay^2+3xy^2)dxdy$$

$$=2c^2\int_0^a\int_0^{x/\sqrt{3}}\left[(2ax-3x^2+3y^2)^2+36y^2(x-a)^2\right]dy\,dx$$

$$-4c\int_0^a\int_0^{x/\sqrt{3}}(ax^2-x^3-3ay^2+3xy^2)dydx$$

$$=\frac{5a^5}{15\sqrt{3}}(2c^2a-c)$$

$$\frac{dJ}{dc}=0\Rightarrow c=\frac{1}{4a}$$

$$u(x,y)=\frac{1}{4a}(x^2-3y^2)(a-x)$$

Incidentally this solution is the exact solution.

Example 7: Find an estimate of the least eigen value of $u_{xx}+u_{yy}+\lambda u=0$ in the region R bounded by the circle $x^2+y^2=1$ and $u=0$ on the boundary.

Here $J=\iint_R(u_x^2+u_y^2-\lambda u^2)dxdy$

Choosing $u=c\left(1-x^2-y^2\right)$ we find

$$\lambda=\frac{\iint_R(x^2+y^2)dxdy}{\iint_R(1-x^2-y^2)^2dxdy}=\frac{4\int_0^1 r^3dr}{\int_0^1(r-2r^3-r^5)dr}=\frac{4\times\frac{1}{4}}{\frac{1}{6}}=6$$

Put Your Own Notes

Example 8: Find an estimate of the least eigen value of $u_{xx} + u_{yy} + \lambda u = 0$

In the isosceles right angled triangular region bounded by the coordinate axes and the line $x + y = 1$. Assume that u vanishes on the boundary of the triangle.

Here $J = \iint_R (u_x^2 + u_y^2 - \lambda u^2)\,dx\,dy$

Choosing, $u(x,y) = cxy(1-x-y)$ we find $u_x = c\left(y - 2xy - y^2\right)$

$u_y = c\left(x - 2xy - x^2\right)$

$\therefore \lambda = \dfrac{I_1}{I_2}$

Where $I_1 = \iint_R (y - 2xy - y^2)^2 + (x - 2xy - x^2)^2\,dxdy$

$I_2 = \iint_R x^2 y^2 (1-x-y)^2\,dxdy$

These integrals are evaluated using the formula $\iint_R f(x,y)\,dxdy = \int_0^1 \int_0^{1-x} f(x,y)\,dy\,dx$

$\int_0^1 x^m (1-x)^n\,dx = \dfrac{m!n!}{(m+n+1)!}$

Thus we obtain $\lambda = \dfrac{8 \times 7!}{6!} = 56$ as an estimate of the smallest eigen value.

The true value is $5\pi^2 = 49.35$.

Direct Substitution Method

$u_{xx} \to -u_x^2$

$u_{yy} \to -u_y^2$

$u \to u^2$

u_x or $u_y \to 0$

Constant or some function of $(x, y) \to$ multiply by $2u$.

Example: Find the approximate solution of poisson equation

$u_{xx} + u_{yy} = -1,\ |x| \le 1,\ |y| \le 1$

$u = 0$ at $x = \pm 1$

and $y = \pm 1$

Solution: By direct substitution method:-

$$-u_x^2 - u_y^2 + 2u = 0 \qquad \ldots(1)$$

Put Your Own Notes

$$I=\int_{-1}^{1}\int_{-1}^{1}\left(u_x^2+u_y^2-2u\right)dx\,dy$$

Put $u=c\left(1-x^2\right)\left(1-y^2\right)$ ...(2)

$$u_x=c(-2x)\left(1-y^2\right),\ u_y=c\left(1-x^2\right)(-2y)$$

$$I=\int_{-1}^{1}\int_{-1}^{1}\left[c^2\left(4x^2\right)\left(1-y^2\right)^2+c^2\left(4y^2\right)\left(1-x^2\right)^2-2c\left(1-y^2-x^2+x^2y^2\right)\right]dx\,dy$$

$$I=\int_{-1}^{1}\int_{-1}^{1}\left[4c^2x^2\left(1-2y^2+y^4\right)+4c^2y^2\left(1-2x^2+x^4\right)-2c\left(1-x^2-y^2+x^2y^2\right)\right]dx\,dy$$

$$=\int_{-1}^{1}\int_{-1}^{1}\left[\left(4c^2x^2-8c^2x^2y^2+4c^2x^2y^4+4c^2y^2-8c^2y^2x^2+4c^2y^2x^4\right)\right.$$

$$\left.-\left(2c-2cx^2-2cy^2+2cy^2x^2\right)\right]dx\,dy$$

$$=\int_{-1}^{1}\int_{-1}^{1}\left[4c^2x^2+4c^2y^2-16c^2x^2y^2+4c^2x^2y^4+4c^2y^2x^4\right.$$

$$\left.-2c+2cx^2+2cy^2-2cx^2y^2\right]dx\,dy$$

$$=\int_{-1}^{1}\left[4c^2x^2y+4c^2\frac{y^3}{3}-16c^2x^2\frac{y^3}{3}+4c^2x^2\frac{y^5}{5}+4c^2\frac{y^3}{3}x^4\right.$$

$$\left.-2cy+2cx^2y+2c\frac{y^3}{3}-2cx^2\frac{y^3}{3}\right]_{-1}^{1}dx$$

$$=\int_{-1}^{1}\left[4c^2x^2(1+1)+4\frac{c^2}{3}(1+1)-16\frac{c^2}{3}x^2(1+1)+4\frac{c^2}{5}x^2(1+1)+4\frac{c^2}{3}x^4(1+1)\right.$$

$$\left.-2c(2)+2cx^2(1+1)+2\frac{c}{3}(1+1)-2\frac{c}{3}x^2(1+1)\right]dx$$

$$=\frac{32}{3}c^2-\frac{64}{9}c^2+\frac{32}{15}c^2+\frac{16}{3}c-\frac{8}{9}c-8c$$

$$\frac{dI}{dc}=0$$

$$\Rightarrow c=\frac{5}{16}$$

Putting $c=\frac{5}{16}$ in equation (2)

$$u(x)=\frac{5}{16}\left(1-x^2\right)\left(1-y^2\right)$$

Put Your Own Notes

1.21. Essential and Suppressible Boundary Conditions

Consider the boundary value problem

$$\frac{d}{dx}(p(x)y')+q(x)=r(x) \qquad \ldots(1)$$

$$\alpha_0\, y(a)+\alpha_1\, y'(a)=\alpha_2 \qquad \ldots(2)$$

$$\beta_0\, y(b)+\beta_1\, y'(b)=\beta_2 \qquad \ldots(3)$$

Let us consider the variational problem of optimizing the functional

$$I=\int_a^b\left(py'^2+qy^2-2ry\right)dx$$

$$\delta I=\int_a^b\delta\left(py'^2+qy^2-2ry\right)dx$$

$$=2\int_a^b\left[py'\delta y'+(qy-r)\delta y\right]dx$$

$$=2\,p\,y'\,\delta\, y\int_a^b dx+2\int_a^b\left[(-py')+qy-r\right]\delta y\, dx$$

$$=2p(b)\,y'(b)\,\delta\, y(b)-2p(a)\,y'(a)\,\delta\, y(a)+2\int_a^b\left[(-py)'+qy-r\right]\delta y\, dx$$

Substituting the values of $y'(a)$ and $y'(b)$ obtained from (2) and (3) respectively, we get

$$\delta I=\frac{2p(b)}{\beta_1}[\beta_2-\beta_0 y(b)]\delta y(b)-\frac{2p(a)}{\alpha_1}[\alpha_2-\alpha_0 y(a)]\delta y(a)+2\int_a^b\left[(-py)'+qy-r\right]\delta y dx$$

$$=\delta\left\{\frac{p(b)}{\beta_1}2\beta_2 y(b)-\beta_0 y^2(b)\right\}-\delta\left\{\frac{p(a)}{\alpha_1}\left(2\alpha_2 y(a)-\alpha_0 y^2(a)\right)\right\}+2\int_a^b\left[(-py')+qy-r\right]\delta y dx$$

Or $\delta J=2\int_a^b\left[(-py')'+qy-r\right]\delta y\, dx$

Where $J=\int_a^b\ (py'^2+qy^2-2ry)dx+\dfrac{p(b)}{\beta_1}\left(\beta_0 y^2(b)-2\beta_2 y(b)\right)$

$$-\frac{p(a)}{\alpha_1}\left(\alpha_0 y^2(a)-2\alpha_2 y(a)\right) \qquad (5)$$

Hence, in order to solve the boundary value problem (1)-(3), we optimize the functional (5), note the functional (4). As we have incorporated the boundary conditions through the extra terms in (5), there is no need for us to see that the choice functions $y_k(x), k=1,2,3,\ldots,n$ in the solution.

$$y(x)=y_0(x)+c_1y_1(x)+c_2y_2(x)+\ldots..+c_ny_n(x)$$

Satisfy the given boundary conditions .Hence the boundary conditions are called suppressible boundary conditions whereas the boundary condition with prescribed values of y are called essential boundary conditions and our choice functions have to satisfy the essential boundary conditions. If $\beta_1=0$ in (3), then the second term involving $p(b)$ in (5) has to be dropped. If $\alpha_1=0$, in (2), then the third term involving p(a) in (5) has to be dropped.

Put Your Own Notes

Example 1: Solve $y''+y=0\ \ y(0)=1,\ y'(1)=0$

The second boundary condition is suppressible and we try the function $y(x)=1+cx$ which satisfies the essential boundary condition. The variation formulation is to optimize.

$$J=\int_0^1\left(y'^2-y^2\right)dx$$

Substituting $y=1+cx$, we get

$$J=\int_0^1\left[c^2-(1+cx)^2\right]dx=\frac{2c^2}{3}-c-1$$

$$\frac{dJ}{dc}=0\Rightarrow c=\frac{3}{4}$$

The approximate solution is $y(x)=1+\frac{3}{4}x$

Comparing this solution with the exact solution, $y(x)=\frac{\cos(1-x)}{\cos 1}$

These two solutions assume the values of 1.750 and 1.851 respectively when $x=1$. Let us choose the two term approximation $y(x)=1+c_1x+c_2x^2$

Now, we find $J=\int_0^1\left[(c_1+2c_2x)^2-\left(1+c_1x+c_2x^2\right)^2\right]dx$

$$=c_1^2-1+2c_1c_2-c_1+\frac{4c_2^2-2c_2-c_1^2}{3}-\frac{c_1c_2}{2}-\frac{c_2^2}{5}$$

$\frac{\partial J}{\partial c_1}=0,\frac{\partial J}{\partial c_2}=0$ yield $\frac{4}{3}c_1+\frac{3}{2}c_2=1$

$$\frac{3}{2}c_1+\frac{34}{15}c_2=\frac{2}{3}$$

Solving, $c_1=\frac{228}{139}$, $c_2=-\frac{110}{139}$

Giving the improved solution $y(x)=1+\frac{228x-110x^2}{139}$

From which we obtain $y(1)=0.849$ which is very close to the exact value 0.851.

Example 2: Estimate the least eigen value of $y''+\lambda y=0$, $y'(0)=0,\ \ y(1)=0$

Here $J=\int_0^1\left(y'^2-\lambda y^2\right)dx$

Choosing one term approximation $y(x)=c(x-1)$, we get

$$J=\int_0^1\left[c^2-\lambda c^2(x-1)^2\right]dx$$

Put Your Own Notes

$$\therefore\ \lambda = \frac{1}{\int_0^1 (x-1)^2\, dx} = 3$$

To improve the estimate, we select a two – term approximation

$$y(x) = (x-1)(c_1 + c_2 x)$$

Now,

$$J = \int_0^1 \Big[(c_1-c_2)^2 + 4c_2^2x^2 + 4c_1(c_1-c_2)x - \lambda\Big\{c_1^2 + (c_1-c_2)x^2 + c_2^2x^4$$

$$-2c_1(c_1-c_2)x + x2c_2(c_1-c_2)x^3 - 2c_1c_2x^2\Big\}\Big]dx$$

$$= (c_1-c_2)^2 + \frac{4c_2^2}{3} + 2c_2(c_1-c_2) - \lambda\Bigg[c_1^2 + \frac{(c_1-c_2)^2}{3} + \frac{c_2^2}{5}$$

$$-c_1(c_1-c_2) + \frac{c_2}{2}(c_1-c_2) - 2\frac{c_1c_2}{3}\Bigg]$$

$$\frac{\partial J}{\partial c_1} = 0, \frac{\partial J}{\partial c_2} = 0 \text{ Yield}$$

$$\frac{2}{3}(3-\lambda)c_1 - \frac{\lambda}{6}c_2 = 0$$

$$-\frac{\lambda}{6}c_1 + \frac{(20-\lambda)}{30}c_2 = 0$$

For non-trivial solution, we must have

$$\begin{vmatrix} \frac{2}{3}(3-\lambda) & -\frac{\lambda}{6} \\ -\frac{\lambda}{6} & \frac{20-\lambda}{30} \end{vmatrix} = 0$$

$$\lambda^2 + 92\lambda - 240 = 0$$

The positive roots is $\lambda = \frac{-92 + \sqrt{92^2 + 4\times 240}}{2}$

or $\lambda = -46 + \sqrt{2356} = 2.539$

Which is nearly equal to the true value $\frac{\pi^2}{4} = 2.467$

Note that the error in the value predicted by one term approximation is 21.6%

Whereas the error in the value predicted by the two – term approximation is only 2.9%.

Example 3: Solve $y'' = 0, y(0) = 1,\ y(1) + y'(1) = 0$

The appropriate variation problem is to optimize the functional

$$J = \int_0^1 y'^2 dx + y^2(1)$$

Put Your Own Notes

Wherein the suppressible boundary condition has been incorporated in J. To satisfy the essential boundary condition $y(0)=1$, we choose $y(x)=1+cx$.

Now, we get

$$J=\int_0^1\left[c^2+(1+c)^2\right]dx=2c^2+2c+1$$

$$\frac{\partial J}{dc}=0\Rightarrow c=-\frac{1}{2}$$

Giving the solution

$$y(x)=1-\frac{x}{2}$$

Which is also the exact solution.

Example 4: Solve $y''+y=x,\ y(0)=0,\ y(1)-y'(1)=0$

Here $J=\int_0^1(y'^2-y^2+2xy)dx-y^2$ (1)

Choosing $y(x)=cx$, which satisfies the essential boundary condition at $x=0$, we find

$$J=\int_0^1(c^2-c^2x^2+2cx^2)dx-c^2$$

$$=\frac{2c-c^2}{3}$$

$$\frac{dJ}{dc}=0\Rightarrow c=1$$

The required solution is $y=x$, which is the exact solution

Example 5: Solve $y''+y=x,\ y'(0)=1,\ y(1)=1$

Here $J=\int_0^1(y'^2-y^2+2xy)dx+2y(0)$

Choosing $y(x)=a+(1-a)x$, we find

$$J=\int_0^1\left[(1-a)^2-a^2-(1-a)^2x^2-2a(1-a)x+2ax+2(1-a)x^2\right]dx+2a$$

$$\frac{dJ}{da}=0\Rightarrow\frac{4a}{3}=0\Rightarrow a=0$$

$\therefore$ The required solution is $y(x)=x$, which is the exact solution.

Exercise: Find the approximate solutions of the following boundary value problems.

(i) $y''+y=1,\ y'(0)=0,\ y(1)=0$

(ii) $y''+y=1,\ y(0)=0,\ y'(1)=0$

(iii) $y''+y=1,\ y(0)=0,\ y'(1)=0$

(iv) $y''+y=x,\ y(0)=0,\ y'(1)=0$

Put Your Own Notes

(v) $y''+\lambda xy=0,\ y(0)=y(1)=0$

(vi) $y''+y=x,\ y(0)=0,\ y'(1)=0$

(vii) $y''+\lambda y=0,\ y(0)=0,\ y(1)+y'(1)=0$

Answer

Exact Solutions

(i) $y(x)=1-\dfrac{\cos x}{\cos 1}$

(ii) $y(x)=1-\dfrac{\cos(1-x)}{\cos 1}$

(iii) $y(x)=1$

(iv) $y(x)=x-\dfrac{\sin x}{\cos 1}$

(v) $\lambda=2$ for one term approximation and $\lambda=19.2$ for two term approximation while exact value is $\lambda=18.9$

(vi) $y(x)=x$

(vii) The eigen value $\lambda=6$ for one term approximation and $\lambda=4.15$ for two term approximation. Exact value is $\lambda=4.12$.

1.22. Proper Field, Central Field and Field of Extremals

Proper Field

A family of curves $y=y(x,c)$ is said to form a proper field in a given region D of the xy plane if one and only one curve of the family passes through every point of the region D. for example, inside the circle $x^2+y^2=1$, the family of parallel lines $y=x+c$ (c being an arbitrary constant) forms a proper field since through any point of the above circle there passes one and only one straight line of the family as shown in figure (i). On the other hand, the family of parabolas $y=(x-c)^2-1$ inside the same circle $x^2+y^2=1$ does not form a proper field since the parabolas of this family intersect inside the circle as shown in figure (ii).

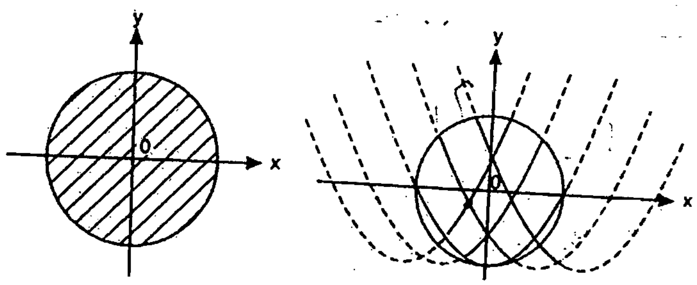

Fig. (i) Fig. (ii)

Central Field

If all the curve of the family $y=y(x,c)$ pass through a certain point (x_0,y_0) *i.e.*, if they form a pencil of curve, then they do not form a proper field in the region D, if the centre of the pencil (x_0, y_0) belongs to D. However, if the curves of the pencil cover the entire region D and do not intersect anywhere in this region, with the exception of the centre of the pencil (x_0, y_0), then family $y = y(x,c)$ is said to form a central field as shown in figure (iii.)

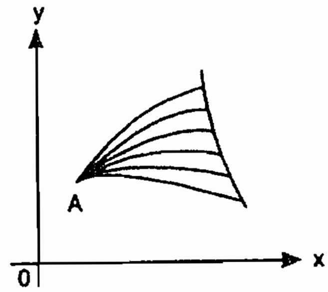

Fig. (iii)

Working rule for testing a weak minimum/weak maximum and strong minimum/strong maximum of an extremal of a given function with help of Legendra condition.

Let $I[y(x)] = \int_{x_1}^{x_2} F(x, y, y')\,dx$ (1)

be a function with fixed boundaries $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. Let $F(x, y, y')$ posses a continuous partial derivative $F_{y'y'}$ that is $\frac{\partial^2 F}{\partial y'^2}$. Let C be the curve of the extremal of the given functional which passes through P_1 and P_2. Also assume that the external of the curve C is included in a field of extremals. Then the following table provides the sufficient conditions for the nature of the extremal of the given functional.

S. No.	Nature of the external of the given function	Sufficient conditions
1.	For a weak minimum to be attained on the extremal on the curve C	$F_{y'y'} > 0$ *on* C
2.	For a weak maximum to be attained on the extremal on the curve C	$F_{y'y'} < 0$ *on* C
3.	For a strong minimum to be attained on the extremal on the curve C	$F_{y'y'} \geq 0$ at points close to C and also for arbitrary values of y'
4.	For a strong maximum to be attained on the extremal on the curve C	$F_{y'y'} \leq 0$ at points close to C and also for arbitrary values of y'

Example: Test for an extermum the function a $I[y(x)] = \int_0^2 (e^{y'} + 3)\,dx, \quad y(0) = 0, \qquad y(2) = 1$

Put Your Own Notes

Solution: Comparing the given function with $\int_0^2 (x,y,y')dx$, *here* $F(x,y,y') = e^{y'} + 3$ ...(1)

Euler's equation is $\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$ ...(2)

From (1), $\frac{\partial F}{\partial y} = 0$, $\frac{\partial F}{\partial y'} = e^{y'}$ so that $\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = \frac{de^{y'}}{dx} = e^{y'} y''$

With these values, (2) gives $0 - e^{y'} y'' = 0$ or $y''=0$

Integrating it, $y' = C_1$ so that $y = C_1 x + C_2$ ...(3)

Showing that an extremal of the given function is attained only on the straight lines (3).

Since $y(0) = 0$ and $y(2) = 1$, (3)

$\Rightarrow 0 = C_2$ and $1=2C_1$ $\Rightarrow$ $C_1 = 1/2, C_2 = 0$

Hence the extermal satisfying the boundary conditions is $y = x/2$ which is included in the central field extremals $y = C_1 x$.

Since $F(x,y,y')$ is three times differentiable with respect to y' for any y' and $F_{y'y'} = e^{y'} > 0$ for any value of y', it follows by (Legendre condition) that on the straight line $y = x/2$ a strong minimum is attained.

ASSIGNMENT SHEET-1

1. The extremal of the functional $\int_0^{\alpha}\left(y'^2-y^2\right)dx$ that passes through $(0,0)$ and $(\alpha,0)$ has a

(a.) weak minimum if $\alpha<\pi$

(b.) strong minimum if $\alpha<\pi$

(c.) weak minimum if $\alpha>\pi$

(d.) strong minimum if $\alpha>\pi$

2. The extremal of the functional $I=\int_0^{x_1} y^2(y')^2\,dx$ that passes through $(0,0)$ and (x_1,y_1) is

(a.) a constant function

(b.) a linear function of x

(c.) part of a parabola

(d.) part of an ellipse

3. Consider the functional

$$J(y)=y^2(1)+\int_0^1 y'^2(x)dx,\ y(0)=1$$

where $y\in C^2([0,1])$ extremizes J then

(a.) $y(x)=1-\frac{1}{2}x^2$

(b.) $y(x)=1-\frac{1}{2}x$

(c.) $y(x)=1+\frac{1}{2}x$

(d.) $y(x)=1+\frac{1}{2}x^2$

4. Let $y\in C^2([0,\pi])$ satisfying $y(0)=y(\pi)=0$ and $\int_0^{\pi}y^2(x)dx=1$ extremize the functional

$$J(y)=\int_0^{\pi}(y'(x))^2dx;\ y'=\frac{dy}{dx}\text{ then}$$

(a.) $y(x)=\sqrt{\frac{2}{\pi}}\sin x$

(b.) $y(x)=-\sqrt{\frac{2}{\pi}}\sin x$

(c.) $y(x)=\sqrt{\frac{2}{\pi}}\cos x$

(d.) $y(x)=-\sqrt{\frac{2}{\pi}}\cos x$

5. The curve extremizing the functional

$$I(y)=\int_1^2\frac{\sqrt{1+(y'(x))^2}}{x}dx,$$

$y(1)=0, y(2)=1$ is

(a.) An ellipse

(b.) A parabola

(c.) A circle

(d.) A straight line

6. Let $u(x,y)$ be an extremal of the functional $J(u)=\iint_D\left[\frac{1}{2}u_x^2+\frac{1}{2}u_y^2+e^{xy}u\right]dxdy$ where D is the open unit disk in R^2. Then u satisfies

(a.) $u_{xx}+u_{yy}-e^{x+y}=0$

(b.) $u_{xx}+u_{yy}=e^{xy}$

(c.) $u_{xx}+u_{yy}=-e^{xy}$

(d.) $\iint_D\left[u_{xx}+u_{yy}-e^{xy}\right]h(x,y)dxdy=0$ for every smooth h vanishing on the boundary of D

7. Consider the functional

$$J(y)=\int_a^b F(x,y,y')dx$$

where $F(x,y,y')=y'+y$ for admissible functions y. Then J has

(a.) no extremals.

(b.) several extremals.

(c.) $y(x)=e^{-x}$ as an extremal.

(d.) $y(x)=$ constant as an extremal.

8. An extremal of the functional

$$J(y)=\int_a^b\sqrt{1+|y'(x)|^2}\,dx$$

(a.) is the straight line connecting $(a,y(a))$ and $(b,y(b))$.

(b.) is a solution to the differential equation $y'=C\sqrt{1+y'^2}$ for some constant C.

(c.) is a solution to $y''=0$.

(d.) does not exist.

9. Consider the functional

$$I(y)=\int_a^b F(y,y')dx;\ y'\equiv\frac{dy}{dx},$$
$$y(a)=y_1,\ \ y(b)=y_2$$

Where $y\in C^2[a,b]$, F has continuous partial derivatives with respect to y, y', and y_1, y_2 are given real numbers. Let $y=y(x)$ be an extremizing function for the functional I. Then along the extremizing curve

(a.) F remains constant

(b.) $\frac{\partial F}{\partial y}=0$

(c.) $F-y\frac{\partial F}{\partial y'}=$ constant

(d.) $F-y'\frac{\partial F}{\partial y'}=$ constant

10. Let $y_{app}(x)$ be a polynomial approximation, involving only one coordinate function, for the functional

$$I(y)=\int_0^1\left(\frac{1}{2}y'^2-y\right)dx;\ y(0)=0, y(1)=0,$$

using Rayleigh-Ritz method; here $y\in C^2[0,1]$. If $y_e(x)$ is an exact extremizing function, then y_e and y_{app} are coincident at

(a.) $x=0$ but not at remaining point [0, 1]

(b.) $x=1$ but not at remaining points in [0, 1]

(c.) $x=0$, $x=1$ but not at other points in [0, 1]

(d.) All point $x\in[0,1]$

11. The extremal of $\int_1^2\frac{\dot{x}^2}{t^3}dt; x(1)=3,\ x(2)=18$ (where $\dot{x}\equiv\frac{dx}{dt}$) using Lagrange's equation is given by which of the following?

(a.) $x=t^4+2$

(b.) $x=\frac{15}{7}t^3+\frac{6}{7}$

(c.) $x=5t^2-2$

(d.) $x=5t^3+3$

12. Let $J(u)=\int_0^1\left[u_x^2+4\frac{u^2}{x^2}\right]x\,dx$, where $u(x)$ is a smooth function defined on $[0,1]$ satisfying $u(0)=0$ and $u(1)=1$. Which of the functions minimizes J?

(a.) $u(x)=x^2$

(b.) $u(x)=\frac{1}{\sqrt{2}}x^2$

(c.) $u(x)=\frac{1}{2}x^2$

(d.) $u(x)=\frac{1}{4}x^2$

13. Consider the functional $J=\int_a^b F(x,y,y')dx$ where $F(x,y,y')=(1+y^2)/y'^2$ for admissible function $y(x)$. Which of the following are extremals for J?

(a.) $y(x)=A\sin(x)$

(b.) $y(x)=A\sinh(x)+B\cosh(x)$

(c.) $y(x)=A\sinh(Ax+B)$

(d.) $y(x)=A\sin(x)+B\cos(x)$.

14. The variation problem of extremizing the functional

$$I(y(x))=\int_1^3 y(3x-y)dx;\ y(3)=4\frac{1}{2},\ y(1)=1$$

has

(a.) A unique solution

(b.) Exactly two solutions

(c.) An infinite number of solutions

(d.) No solution.

15. An approximate solution $z=z_\circ(x,y)$ to the problem of extremizing the function

$$f(z(x,y))=\iint_D\left[\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2-2z\right]dxdy$$

where D is the square. $-1\le x\le 1, -1\le y\le 1$ and $z=0$ on the boundary of the square, is of the form (a.) $z_0 = \sum_{i=1}^{n} a_i \phi_i(x,y)$, where a_i are constants and functions ϕ_i are linearly independent in D.

(b.) $z_\circ = a_1 \phi_1(x,y) + a_2 \phi_2(x,y)$ where a_1 and a_2 are constants and ϕ_1 and ϕ_2 have continuous partial derivatives.

(c.) $z_\circ = a\phi(x,y)$ where α is a constant and ϕ is continuous in D

(d.) $z_\circ = (x^2-1)(y^2-1)/16$

16. Consider the functional $I(y) = \int_{1/2}^{x_1} \sqrt{1+y'^2}\,dx$ subject to the conditions that the left end of the extremal is fixed at the point $\left(\frac{1}{2}, \frac{1}{4}\right)$ and the right end (x_1, y_1) be movable along the straight line $y = x-5$. Let C_1 and C_2 be arbitrary constants. Then

(a.) The general solution of the Euler's equation satisfies $2C_1 + 4C_2 = 1$

(b.) The transversality condition that holds at the right end yields $\sqrt{(1+C_1^2)} + (1-C_1)\frac{C_1}{\sqrt{1+C_1^2}} = 0$

(c.) The relation $C_1 x_1 + C_2 = x_1 - 5$ is true

(d.) The extremal is $y = -x + \frac{3}{4}$

17. The variational problem of extremizing the functional

$$I(y(x)) = \int_0^{2\pi} \left[\left(\frac{d}{dx}y\right)^2 - y^2\right] dx;\ y(0)=1,\ y(2\pi)=1$$

has

(a.) A unique solution

(b.) Exactly two solutions

(c.) An infinite number of solutions

(d.) No solution

18. In the Ritz method, seeking an extremum of the functional

$$I(y) = \int_{x_0}^{x_1} F\left(x, y, \frac{dy}{dx}\right) dx;\quad y(x_0)=a, y(x_1)=b.$$

The coordinate function/or the admissible function $\phi_i(x), i = 1, 2, ...$ defined on $[x_0,\ x_1]$ must be

(a.) Linearly independent

(b.) Continuous

(c.) Smooth

(d.) Linearly independent, smooth and the functional be considered not along admissible curves $y = y(x)$ but only along all possible linear combinations of admissible functions.

19. The extremizing curve for the functional $\int_{x_0}^{x_1} F(y', z')\,dx$ with $F_{y'y'}F_{z'z'} - (F_{y'z'})^2 \neq 0$,

(a.) Is uniquely determined

(b.) Is a member of an infinite family of curves laying in a plane

(c.) Belongs to a family of curves in space

(d.) Does not exist.

20. An extremum of the functional $\int_0^1 \left[(y')^2 + 12xy\right] dx,\ y(0)=0, y(1)=1$ can occur only along the curve

(a.) $x(1-x)e^x$

(b.) $(1-x)x^2$

(c.) x^2

(d.) x^3

21. Any function that gives an extremum of the functional

$$\iint_D \left\{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 2zf(x,y)\right\} dx\,dy$$

satisfy the so called

(a.) Harmonic equation

(b.) Heat equation

(c.) Wave equation

(d.) Poisson equation

22. Let $J[y] = \int_{-1}^{1} (y'e^y + xy^2)\,dx$ be a functional defined on $C^1[-1,1]$. Then the variation of $J[y]$ is (a.) $\int_{-1}^{1}\left(y''e^{y}+e^{y}y'+2xyy'\right)dx$

(b.) $\int_{-1}^{1}\left(y'^{2}e^{y}+y^{2}+2xyy'\right)dx$

(c.) $\int_{-1}^{1}\left[\left(y'e^{y}+2xy\right)\delta y+e^{y}\delta y'\right]dx$

(d.) $\int_{-1}^{1}\left[\left(y'e^{y}+2xy+y^{2}\right)\delta y+e^{y}\delta y'\right]dx$

23. Consider the functional $J[y]=\int_{1}^{2}\left(xy'^{4}-2yy'^{3}\right)dx$ defined on $S=\left\{y/y\in C^{1}[1,2]\, and\ y(1)=0,y(2)=1\right\}$ Then the functional attains

(a.) A weak minimum on $y=x-1$

(b.) A strong minimum on $y=x-1$

(c.) A weak maximum on $y=x-1$

(d.) A strong maximum on $y=x-1$

24. The extremal problem $J[y(x)]=\int_{0}^{\pi}\left\{(y')^{2}-y^{2}\right\}dx$, $y(0)=1,y(\pi)=\lambda$, has

(a.) A unique extremal if $\lambda=1$

(b.) Infinitely many extremals if $\lambda=1$

(c.) A unique extremal if $\lambda=-1$

(d.) Infinitely many extremal if $\lambda=-1$

25. The functional $J[y]=\int_{0}^{1}e^{x}\left(y^{2}+\frac{1}{2}y'^{2}\right)dx;\ y(0)=1,y(1)=e$ attains

(a.) A weak, but not a strong minimum on e^{x}

(b.) A strong minimum on e^{x}

(c.) A weak, but not a strong maximum on e^{x}

(d.) A strong maximum on e^{x}

26. Consider the functional $F(u,v)=\int_{0}^{\pi/2}\left[\left(\frac{\partial u}{\partial x}\right)^{2}+\left(\frac{\partial v}{\partial x}\right)^{2}+2u(x)v(x)\right]dx$ with $u(0)=1,v(0)=-1$ and $u\left(\frac{\pi}{2}\right)=0,v\left(\frac{\pi}{2}\right)=0$. Then, the extremals satisfy

(a.) $u(\pi)=1,v(\pi)=-1$

(b.) $u(\pi)=1,v(\pi)=0,u(\pi)-v(\pi)=2$

(c.) $u(\pi)=-1,v(\pi)=1$

(d.) $u(\pi)+v(\pi)=-2,u(\pi)-v(\pi)=0$

27. If $u(x)$ and $v(x)$ satisfying $u(0)=1,v(0)=-1$, $u\left(\frac{\pi}{2}\right)=0$ and $v\left(\frac{\pi}{2}\right)=0$ are the extremals of the functional $\int_{0}^{\pi/2}\left\{\left(\frac{du}{dx}\right)^{2}+\left(\frac{dv}{dx}\right)^{2}+2uv\right\}dx$, then

(a.) $u\left(\frac{\pi}{4}\right)+v\left(\frac{\pi}{4}\right)=0$

(b.) $u\left(\frac{\pi}{3}\right)-v\left(\frac{\pi}{3}\right)=0$

(c.) $u\left(\frac{\pi}{3}\right)+v\left(\frac{\pi}{3}\right)=1$

(d.) $u\left(\frac{\pi}{3}\right)+v\left(\frac{\pi}{3}\right)=0$

28. The functional $\int_{0}^{1}(y'^{2}+(y+2y')y''+kxyy'+y^{2})dx$, $y(0)=0,y(1)=1,y'(0)=2,y'(1)=3$ is path independent if k equals

(a.) 1

(b.) 2

(c.) 3

(d.) 4

29. The functional $\int_0^1 \left(y'^2+4y^2+8ye^x\right)dx$, $y(0)=-\frac{4}{3}$, $y(1)=-\frac{4e}{3}$ possesses:

(a.) Strong minima on $y=-\frac{1}{3}e^x$

(b.) Strong minima on $y=-\frac{4}{3}e^x$

(c.) Weak maxima on $y=-\frac{1}{3}e^x$

(d.) Strong maxima on $y=-\frac{4}{3}e^x$

30. On the interval $[0,1]$, let y be a twice continuously differentiable function which is an extremal of the functional $J(y)=\int_0^1 \frac{\sqrt{1+2y'^2}}{x}dx$ with $y(0)=1, y(1)=2$. Then, for some arbitrary constant c, y satisfies

(a.) $y'^2\left(2-c^2x^2\right)=c^2x^2$

(b.) $y'^2\left(2+c^2x^2\right)=c^2x^2$

(c.) $y'^2\left(1-c^2x^2\right)=c^2x^2$

(d.) None of these

31. The Euler's equation for the variational problem: Minimize $I[y(x)]=\int_0^1 (2x-xy-y')y'dx$, is

(a.) $2y''-y=2$

(b.) $2y''+y=2$

(c.) $y''+2y=0$

(d.) $2y''-y=0$

32. The extremal of the functional $\int_0^1\left(y+x^2+\frac{y'^2}{4}\right)dx, y(0)=0, y(1)=0$ is

(a.) $4\left(x^2-x\right)$

(b.) $3\left(x^2-x\right)$

(c.) $2\left(x^2-x\right)$

(d.) x^2-x

33. The functional $\int_0^1\left(y'^2+x^3\right)dx$, given $y(1)=1$, achieves its

(a.) Weak maximum on all its extremals

(b.) Weak minimum on all its extremals

(c.) Weak maximum on some, but not on all of its extremals

(d.) Weak minimum on some, but not on all of its extremals

34. The possible values of α for which the variational problem $J[y(x)]=\int_0^1\left(3y^2+2x^3y'\right)dx, y(\alpha)=1$ has externals are

(a.) −1, 0

(b.) 0, 1

(c.) −1, 1

(d.) −1, 0, 1

35. The extremum for the variational problem $\int_0^{\frac{\pi}{8}}\left((y')^2+2yy'-16y^2\right)dx,\ y(0)=0,\ y\left(\frac{\pi}{8}\right)=1$ occurs for the curve

(a.) $y=\sin(4x)$

(b.) $y=\sqrt{2}\sin(2x)$

(c.) $y=1-\cos(4x)$

(d.) $y=\frac{1-\cos(8x)}{2}$

36. The functional $\int_0^1(1+x)(y')^2dx, y(0)=0, y(1)=1$, has

(a.) strong maxima

(b.) strong minima

(c.) weak maxima but NOT a strong maxima

(d.) weak minima but NOT a strong minima

37. Let I be the functional defined by $I(y(x))=\int_0^{\pi/2}\left\{\left(\frac{dy}{dx}\right)^2-y^2\right\}dx;\ y(0)=0,$ $y(\pi/2)=1$ where the unknown function $y(x)$ possesses two derivatives everywhere in $(0,\pi/2)$. Then

(a.) The functional has an extremum which cannot be achieved in the class of continuous functions
(b.) The corresponding Euler's equation does not have a unique solution satisfying the given boundary conditions
(c.) I is not linear
(d.) I is linear

38. Let $I(y(x))=\int_0^1 F\left(x,y,\frac{dy}{dx}\right)dx,$ satisfying $y(0)=0,\ y(1)=1,$ where F has continuous second order derivatives with respect to its arguments and the unknown function $y(x)$ possess two derivatives every where in (0,1). If the function F depends only on x and $\frac{dy}{dx}$, then the Euler's equation is an ordinary differential equation in y which, in general, is
(a.) First order linear
(b.) First order nonlinear
(c.) Second order linear
(d.) Second order nonlinear

39. The functional $I(y(x))=\int_0^1\left(y+\frac{d^2y}{dx^2}\right)dx$ defined on the set of functions $C^2([0,1])$ satisfying $y(0)=1,\ y(1)=1,\ \left(\frac{dy}{dx}\right)_{x=0}=0$ and $\left(\frac{dy}{dx}\right)_{x=1}=-1$ has
(a.) Only one extremal
(b.) Exactly two extremals
(c.) Infinite number of extremals
(d.) No extremals

40. The extremum of the functional $I=\int_0^1\left[\left(\frac{dy}{dx}\right)^2+12xy\right]dx$ satisfying the conditions $y(0)=0$ and $y(1)=1$ is attained on the curve
(a.) $y=\sin^2\frac{\pi x}{2}$
(b.) $y=\sin\frac{\pi x}{2}$
(c.) $y=x^3$
(d.) $y=\frac{1}{2}\left[x^3+\sin\frac{\pi x}{2}\right]$

41. The extremals for the functional $v[y(x)]=\int_{x_0}^{x_1}\left(xy'+y'^2\right)dx$ are given by the following family of curves:
(a.) $y=c_1+c_2x+\left(\frac{x^2}{4}\right)$
(b.) $y=1+c_1x+c_2\left(\frac{x^2}{4}\right)$
(c.) $y=c_1+x+c_2\left(\frac{x^4}{4}\right)$
(d.) $y=c_1+c_1x-\left(\frac{x^2}{4}\right)$

42. Extremals for the variational problem $v[y(x)]=\int_1^2\left(y^2+x^2y'^2\right)dx$ satisfy the differential equation
(a.) $x^2y''+2xy'-y=0$
(b.) $x^2y''-2xy'+y=0$
(c.) $2xy'-y=0$
(d.) $x^2y''-y=0$

43. The functional
$$v[y(x)]=\int_0^2\left[(y')^2+6xy+x^3\right]dx,$$
$y(0)=0,\ y(2)=2$ can be extremized on the curve
(a.) $y=x$
(b.) $2y=x^3$
(c.) $y=x^3-6x$
(d.) $2y=x^3-2x$

44. Extremals $y=y(x)$ for the variational problem $v[y(x)]=\int_0^1(y+y')^2dx$ satisfy the differential equation
(a.) $y''+y=0$
(b.) $y''-y=0$
(c.) $y''+y'=0$
(d.) $y'+y=0$

45. An external of the functional $I[y(x)] = \int_a^b F\left(x, y, \frac{dy}{dx}\right)dx; y(a) = y_1, y(b) = y_2$ satisfies Euler's equation, which in general

(a.) Is a second order linear ordinary differential equation (ODE)

(b.) Is a nonlinear ODE of order greater than two

(c.) Admits a unique solution satisfying the conditions $y(a) = y_1, y(b) = y_2$

(d.) May not admit a solution satisfying the conditions $y(a) = y_1, y(b) = y_2$

46. Suppose that among all continuously differentiable functions $y(x), x \in \mathbb{R}$ with $y(0) = 0$ and $y(1) = \frac{1}{2}$, the function $y_0(x)$ minimizes the functional $\int_0^1 \left(e^{-(y'-x)} + (1+y)y'\right)dx$. Then $y_0\left(\frac{1}{2}\right)$ is equal to

(a.) 0

(b.) $\frac{1}{8}$

(c.) $\frac{1}{4}$

(d.) $\frac{1}{2}$

47. Let $y = y(x)$ be the extremal of the functional

$$I[y(x)] = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx,$$

Subject to the condition that the left end of the extremal moves along $y = x^2$, while the right end moves along $x - y = 5$. Then the

(a.) Shortes distance between the parabola and the straight line is $\left(\frac{19\sqrt{2}}{8}\right)$.

(b.) Slope of the extremal at (x, y) is $\left(-\frac{3}{2}\right)$.

(c.) Point $\left(\frac{3}{4}, 0\right)$ lies on the extremal.

(d.) Extremal is orthogonal to the curve $y = \frac{x}{2}$.